THE UNIVERSITY OF DA NANG

VN-UK INSTITUTE FOR RESEARCH AND EXECUTIVE EDUCATION

FINAL EXAMINATION — CREATIVE PROJECT

Introduction to Human–Computer Interaction

Opportunities

A multifunction note-taking app — capture anything, find it in seconds, never lose it

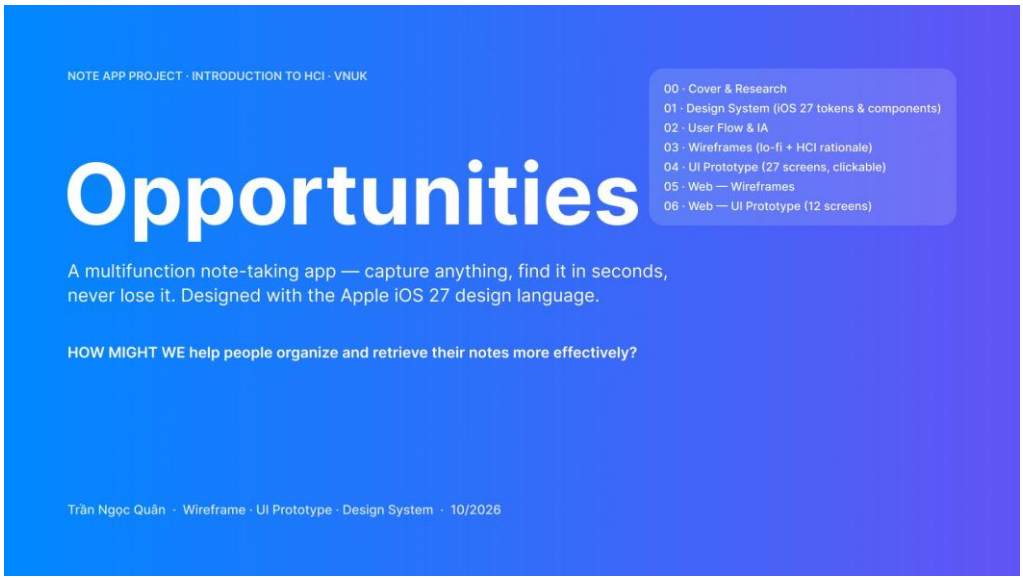


Figure 1. Project cover (Figma)

Student	Trần Ngọc Quân
Student number	……
Class	……
Lecturer	Nguyen Thi Diem Chi (Chi Nguyen)
Module	Introduction to Human–Computer Interaction — Final project (60%)
Submitted	October 2026

TABLE OF CONTENTS

PART 1 – ESSAY 3
I. Brief of the project 3
Background 3
Problem → Solution 4
II. Analyze 5
A. Scenarios 5
B. Persona 5
C. Sympathy map 9
D. Feature list 10
III. Produce 12
A. Flow chart 12
B. UX wireframe 20
IV. Test cases 30
PART 2 – PROTOTYPE 33
Visual design system 33
I. Mobile app 35
II. Web 44

(Right-click the table → Update Field to refresh page numbers.)

PART 1 – ESSAY

I. Brief of the project

Background

- Students and young professionals take notes everywhere: typed lecture notes on a laptop, whiteboard photos on the phone, ideas dictated on the bus, checklists in chat apps and tuition records in a paper notebook.
- Because this information is spread across five or more tools, people waste time searching, forget deadlines, and lose notes when a phone is reset or Wi-Fi is weak.
- Handwritten notes, screenshots and scans cannot be searched, and typing formulas or tables on a phone is painful.
- Existing workspace apps are powerful but feel heavy for casual users, while simple apps lack structure and offline reliability.

How might we *help people organize and retrieve their notes more effectively?*

- **A. Project:** Opportunities — multifunction note-taking app (iOS mobile + desktop web app)
- **B. Brand:** Opportunities — “every note is an opportunity you don’t lose”
- **C. Client:** University students, part-time tutors and young professionals who manage study, work and personal notes
- **D. Project descriptions:**
  - Quick Capture in two taps: text, checklist, whiteboard scan, voice, drawing and table
  - Scan whiteboards with edge detection; on-device text recognition makes photos and handwriting searchable
  - Organize with folders, nested tags, colour labels, pins and [[links]] between notes
  - Checklist items with reminders and one “what’s due” Tasks view across all notes
  - Math palette for formulas (no LaTeX needed) and a Markdown/LaTeX source mode for power users
  - Offline-first auto-save, sync between phone and laptop, version history and a 30-day trash
  - Tutoring templates: lesson plan, attendance and a tuition table with automatic totals
  - Share read-only links, export PDF / Markdown; Dark mode and Focus mode
- **E. Budget:** N/A (academic project)
- **F. Brand background:** Designed by Trần Ngọc Quân for the HCI course at VNUK, 2026. Motto: “Capture anything. Find it in seconds. Never lose it.”
- **G. Target audience:**
  - Primary — busy, mid-tech students who also tutor part-time (persona Trần Minh Thư)

    - Secondary — keyboard-first power users who write Markdown and LaTeX (persona Phạm Đức Anh)
- **H. Coverage:** Vietnamese students and young professionals; UI in Vietnamese with English as an option
- **I. Timing:** Research and analysis → wireframes → hi-fi prototype → heuristic evaluation and v2 re-design (October 2026)
- **J. Description (platform):**
    - Mobile: iPhone 17 Pro (402 × 874 pt) built with the Apple iOS & iPadOS 27 UI kit (Liquid Glass, SF type scale, system colours)
    - Web: desktop web app 1440 × 900, 3-column workspace (sidebar · list · editor) with keyboard shortcuts
    - Tools: Figma (design system, wireframes, prototype), draw.io (diagrams)
- **K. Objective:**
    - Reduce the number of apps users need to one, and let them capture any note in ≤ 2 taps
    - Let users find any note — including text inside photos — in ≤ 2 seconds / ≤ 3 actions
    - Never lose a note: offline-first saving, sync, version history and undo
    - Stay simple for beginners and powerful for experts (Simple ↔ Markdown modes)

Problem → Solution

Problem	Solution — Opportunities	Success metric
Notes are stored in different places; handwriting, screenshots and scans are not searchable; typing is slow when busy; users forget deadlines, lose data, cannot work offline; many apps are too complicated.	One platform for text, voice, photos, scans, drawings, checklists and tables; folders, tags, colours, pins and links; text search inside photos, reminders, auto-save, offline access, phone ↔ web sync, themes, text size and focus mode.	Capture ≤ 2 taps · find a note ≤ 2 s · 0 notes lost · formula inserted without typing code · SUS ≥ 80

II. Analyze

A. Scenarios

Scenario 1 — Trần Minh Thư (primary persona): a Thursday of lectures, tutoring and exam revision

- **Step 1 (07:55, on the bus):** Thư opens Opportunities from the home-screen widget and taps "+" → Voice. She says "Remind Minh to do exercise 2.4, quiz on Thursday". The app transcribes it and shows "I understood: add task, remind Thursday 19:00, save to Tutoring › Minh". She taps Save.
- **Step 2 (09:40, in class):** The lecturer writes on the whiteboard. Thư taps + → Scan. The camera detects the board's edges and captures automatically; a progress card shows "Recognizing text… line 2/4 — works offline".
- **Step 3 (09:42):** On Review Scan she corrects one word, chooses folder Lecture › Giải tích 2 and tag #calculus, then taps Save. The badge shows "Saved on iPhone".
- **Step 4 (09:45):** In the note editor (Live Preview) she needs a formula. She opens the math palette, taps ∫, fills the limits 0 and 2π and taps Insert — no LaTeX typed. She also sketches a graph with Drawing and converts her handwritten "y = x² − 2x" into a formula.
- **Step 5 (12:30, canteen without Wi-Fi):** A banner says "Offline — still saved on this phone; 3 changes will sync when online". She keeps working.
- **Step 6 (17:30):** A lock-screen notification "Collect October fee — Minh, 18:00" appears. She taps "Paid ✓"; the tuition table updates its total automatically.
- **Step 7 (20:30, laptop):** She opens the web app. Because she edited the same note offline, a merge dialog lets her keep lines from both versions — nothing is lost.
- **Step 8 (23:00):** She turns on Dark + Focus mode and searches "tích phân"; results include the text recognized in this morning's whiteboard photo, highlighted.

Scenario 2 — Phạm Đức Anh (power user): writing his thesis note

- **Step 1:** On his MacBook he opens the web app and presses ⌘K (Ctrl K on Windows) to open the command palette, types "ins" and chooses Insert formula.
- **Step 2:** He switches to Source mode (⌘E). Typing \fr shows LaTeX autocomplete; Tab inserts \frac{▢}{▢} with placeholders. The right pane previews the rendered equation (1).
- **Step 3:** He links related notes with [[Bài tập chương 3]]; the backlinks panel shows the notes that mention this one.
- **Step 4:** He right-clicks the note → Export PDF / Markdown (Ctrl ⇧ E) and sends a read-only link to his advisor.

B. Persona

Two personas were built from the user research. Thư is the primary persona (most design decisions are optimised for her); Đức Anh is the power-user persona who validates advanced features.

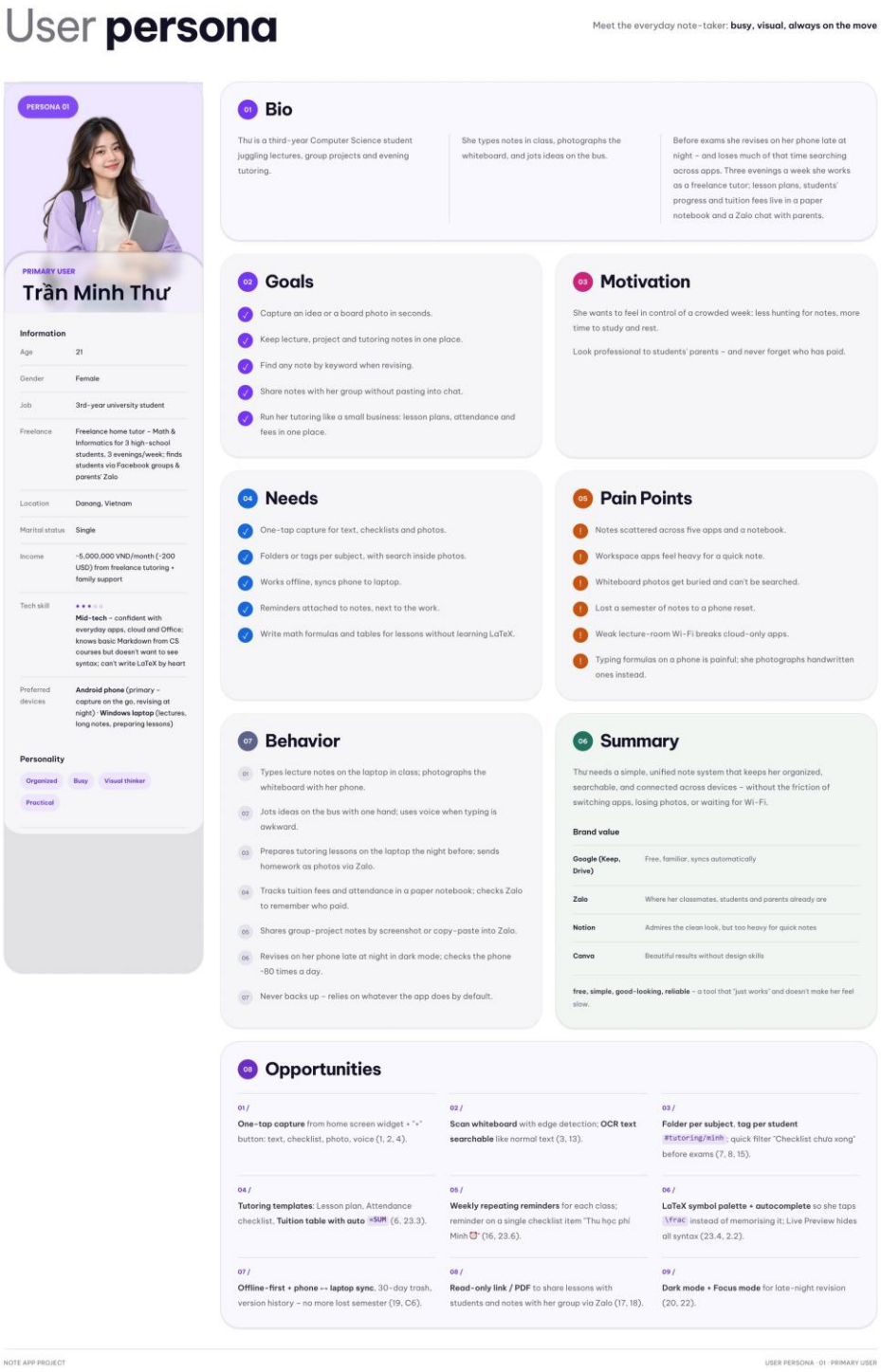


Figure 2. Persona 01 — Trần Minh Thư, primary user (busy, visual, always on the move)

Field	Trần Minh Thư — Persona 01 (Primary)
Overall description	21, female, Danang. 3rd-year Computer Science student and freelance home tutor (Math & Informatics, 3 evenings/week). Mid-tech: confident with everyday apps, knows basic Markdown but does not want to see syntax and cannot write LaTeX by heart. Android phone + Windows laptop.

Field	Trần Minh Thư — Persona 01 (Primary)
Goals	• Capture an idea or board photo in seconds • Keep lecture, project and tutoring notes in one place • Find any note by keyword when revising • Share notes with her group without pasting into chat • Run tutoring like a small business: lesson plans, attendance and fees in one place
Pain points	• Notes scattered across five apps and a notebook • Workspace apps feel heavy for a quick note • Whiteboard photos get buried and can’t be searched • Lost a semester of notes to a phone reset • Weak lecture-room Wi-Fi breaks cloud-only apps • Typing formulas on a phone is painful
Needs	One-tap capture; folders/tags per subject with search inside photos; works offline and syncs; reminders next to the work; formulas and tables without LaTeX

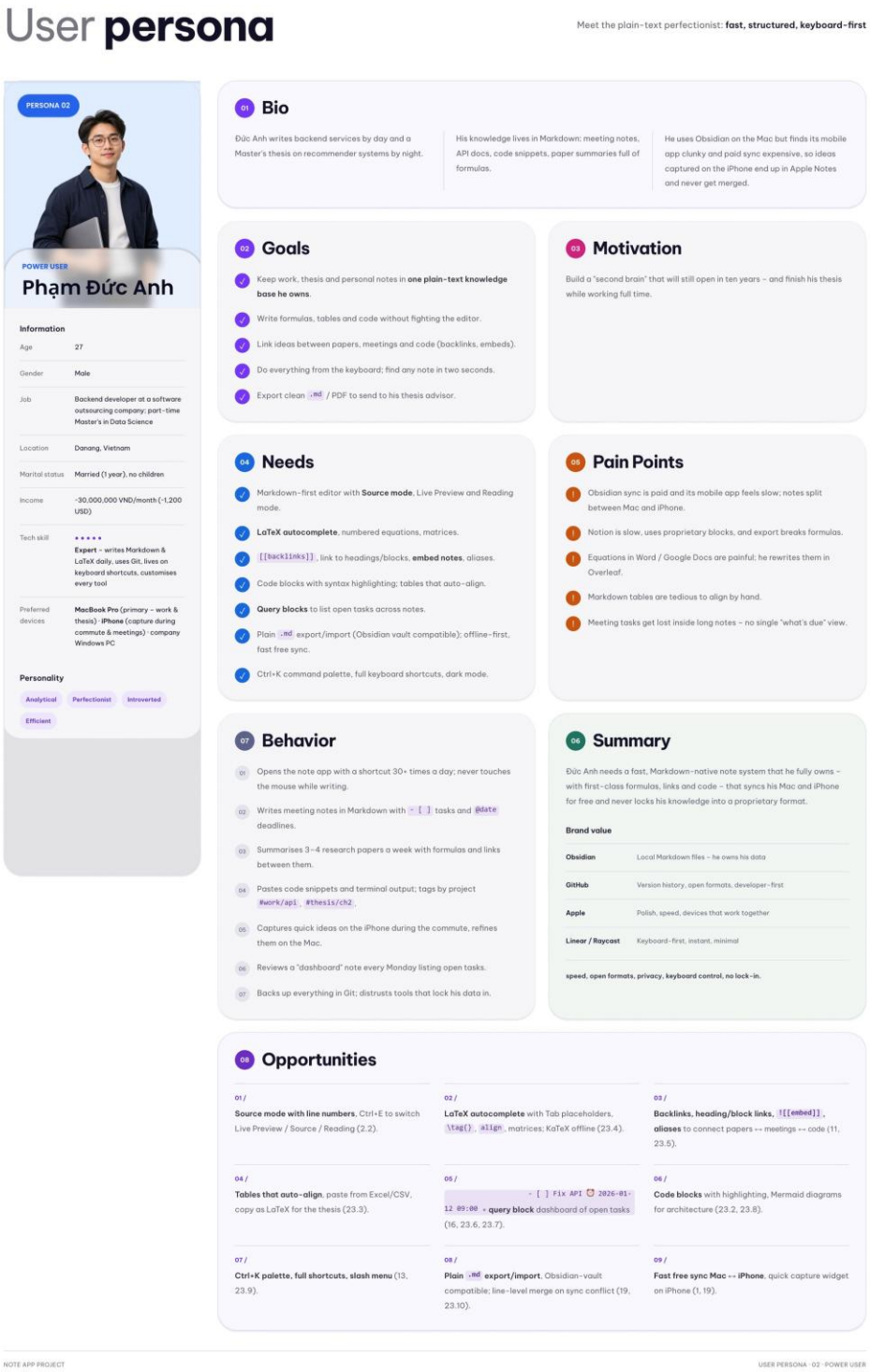


Figure 3. Persona 02 — Phạm Đức Anh, power user (fast, structured, keyboard-first)

Field	Phạm Đức Anh — Persona 02 (Power user)
Overall description	27, male, Danang. Backend developer and part-time Master's student in Data Science. Expert: writes Markdown and LaTeX daily, uses Git, lives on keyboard shortcuts. MacBook Pro + iPhone.

Field	Phạm Đức Anh — Persona 02 (Power user)
Goals	• Keep work, thesis and personal notes in one plain-text knowledge base he owns • Write formulas, tables and code without fighting the editor • Link ideas between papers, meetings and code • Do everything from the keyboard; find any note in two seconds • Export clean .md / PDF for his thesis advisor
Pain points	• Obsidian sync is paid and its mobile app feels slow • Notion is slow and export breaks formulas • Markdown tables are tedious to align by hand • Meeting tasks get lost inside long notes — no single “what’s due” view
Needs	Markdown-first editor with Live Preview / Source / Reading; LaTeX autocomplete; [[backlinks]]; query of open tasks; plain .md export; fast free sync; Ctrl/⌘ K palette and dark mode

C. Sympathy map

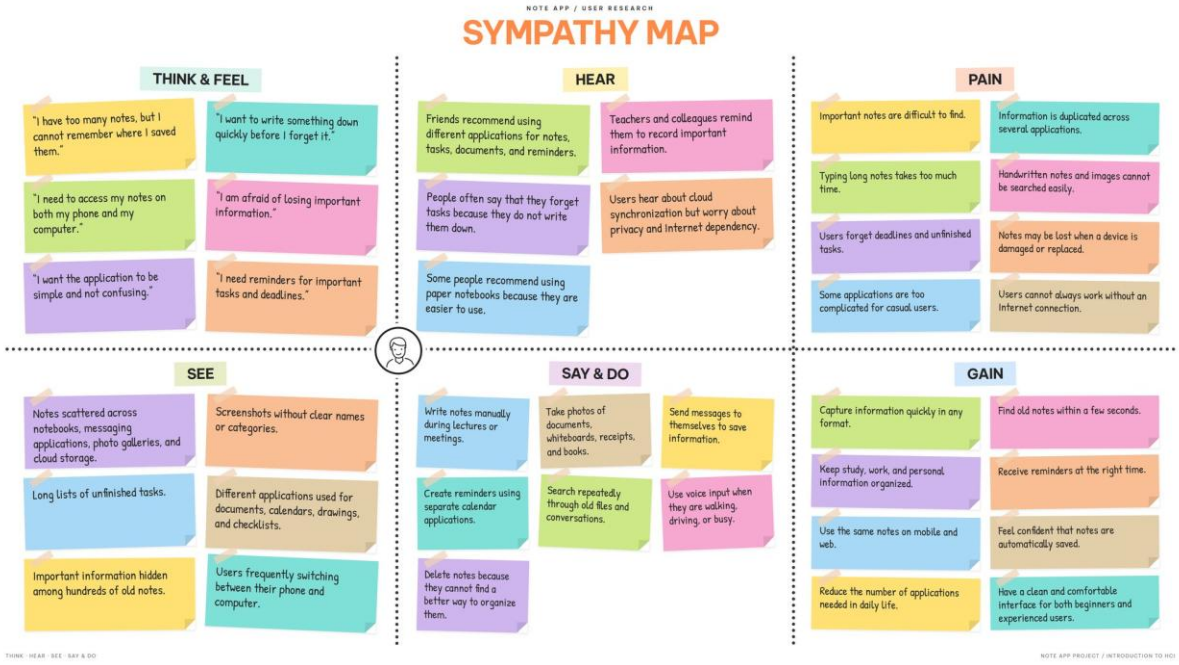


Figure 4. Sympathy map — note app user research (Think & Feel, Hear, See, Say & Do, Pain, Gain)

Quadrant	Key insights → design response
Think & Feel	“I have too many notes but can’t remember where I saved them”, “I want to write something down quickly before I forget”, “I’m afraid of losing important information”, “I want it simple, not confusing” → Quick Capture, search, offline-first + undo, Simple mode
Hear	Friends recommend different apps; teachers remind them to record information; worries about cloud sync and privacy → one app, local-first storage, read-only sharing
See	Notes scattered across notebooks, chat apps, galleries and cloud; screenshots without names; long lists of unfinished tasks → folders, tags, OCR, Tasks view

Quadrant	Key insights → design response
Say & Do	Write manually in lectures, photograph documents and whiteboards, send messages to themselves, search repeatedly through old files, use voice when busy → scan, voice capture, smart filters
Pain	Important notes are hard to find; typing long notes takes time; forget deadlines; handwriting and images can't be searched; notes lost when a device is replaced; can't work without Internet
Gain	Capture in any format; find old notes within seconds; keep study/work/personal organized; reminders at the right time; same notes on mobile and web; confident notes are saved automatically; clean UI for beginners and experts

D. Feature list

a) Functional

- Quick Capture: Create a note in ≤ 2 taps as text, checklist, scan, voice, drawing or table; browse, preview, edit, save and reopen predefined lesson-plan, attendance and tuition-table templates.
- Scan, files & recognition: Document camera with edge detection and auto-capture; visible recognition progress with cancel; editable Vietnamese/English OCR; attachment import, unreadable-file recovery and explicit microphone/camera permission fallbacks.
- Note editor: Live Preview, Source (Markdown + LaTeX autocomplete) and Read-only modes; editor-tools panel; math palette; drawing-to-text/formula; attachments; tables with automatic totals; [[backlinks]].
- Browse & organize: List, grid and calendar views; sorting and note selection; folders per subject, folder appearance, nested #tags, colour labels, pins, smart folders and recoverable folder deletion.
- **Search:** Searches note text and text inside photos/handwriting, with scope filters, highlighted matches and recent searches; works offline.
- Tasks & reminders: Checklist items with date/time, repeat and early reminder; edit/delete task flows; Tasks views (Today, Scheduled, All, Overdue); collapsed/expanded lock-screen notifications with Complete, Snooze and Clear outcomes.
- Share & export: Read-only link with 7-day, custom or no-expiry options; QR sharing; PDF with rendered formulas; Markdown with optional images; share to Zalo, Messages or Mail.
- Sync & safety: Offline-first auto-save; visible syncing/offline/failure states; retry or keep working offline; line-by-line conflict merge; completion feedback; version history; 30-day Recently Deleted; undo and permanent-delete confirmation.
- Comfort: Dark and Focus modes, three text sizes, Simple/Markdown editor style, Vietnamese/English interface and direct access to device permissions.

b) Non-functional

- **Performance:** App opens in < 1 s from local storage; search returns results in ≤ 2 s for 1,000 notes; auto-save within 1 s of typing.

- **Reliability:** All changes are written locally first; no data loss when offline or on conflict; deleted notes are recoverable for 30 days.
- **Usability:** Every touch target ≥ 44 × 44 pt; capture ≤ 2 taps; system status always visible; destructive actions confirmed or undoable.
- **Accessibility:** Dynamic Type text sizes, Dark mode, colour never the only cue (icon + text), keyboard focus states and shortcuts on web.
- **Privacy & security:** Text recognition runs on-device; shared links are read-only and can expire; user owns and can export all data.
- **Consistency:** Built on Apple iOS 27 components and system colours; one icon set; Ctrl on Windows, ⌘ on macOS.

III. Produce

A. Flow chart

The interaction was modelled with nine diagrams (draw.io): use case, sequence, user flow, Norman's interaction model, customer journey map, information architecture, screen flow, state transition and hierarchical task analysis.

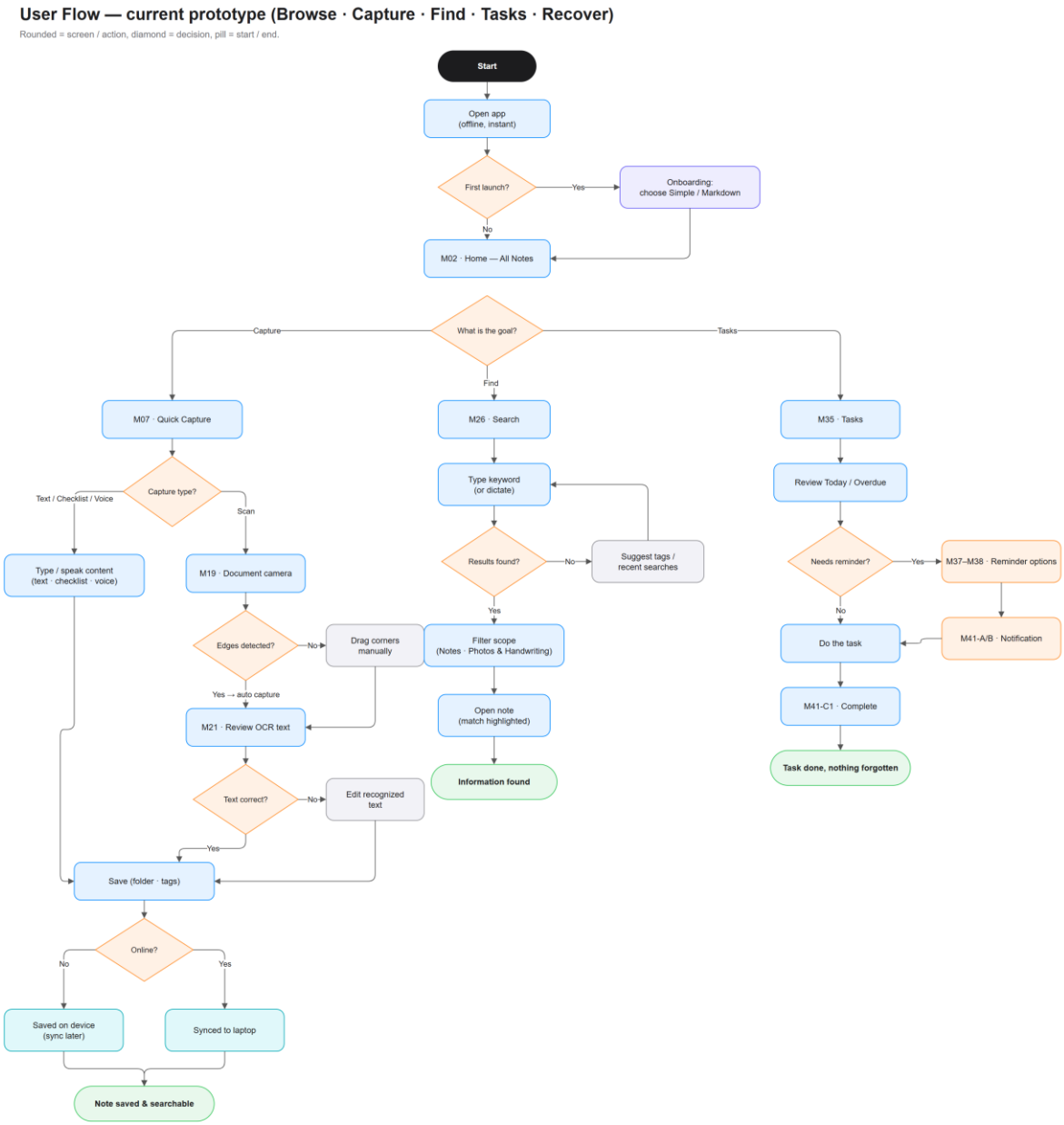


Figure 5. User flow — current prototype: Browse · Capture · Find · Tasks · Recover (rounded = screen/action, diamond = decision)

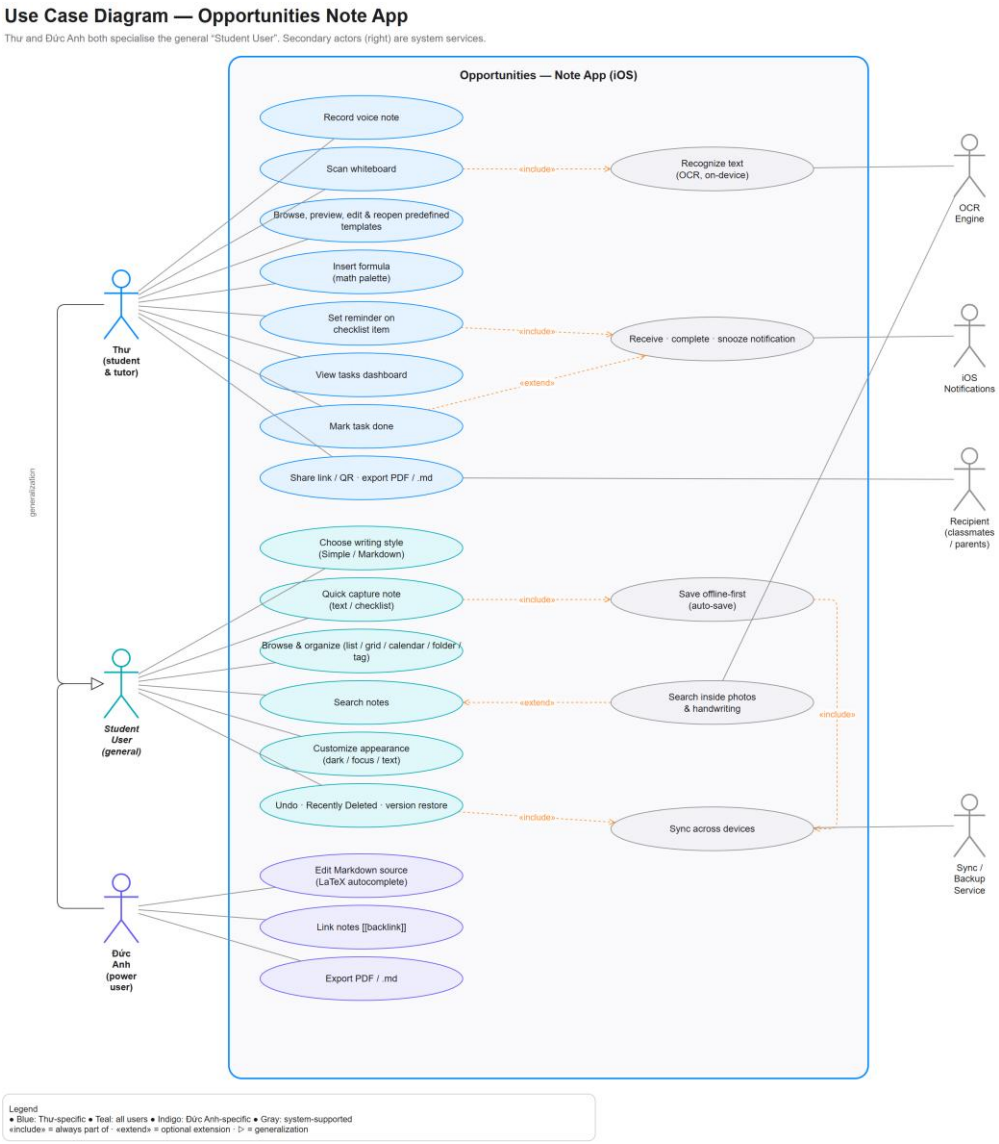


Figure 6. Use case diagram — actors Thư, Đức Anh (specialising Student User) and system services

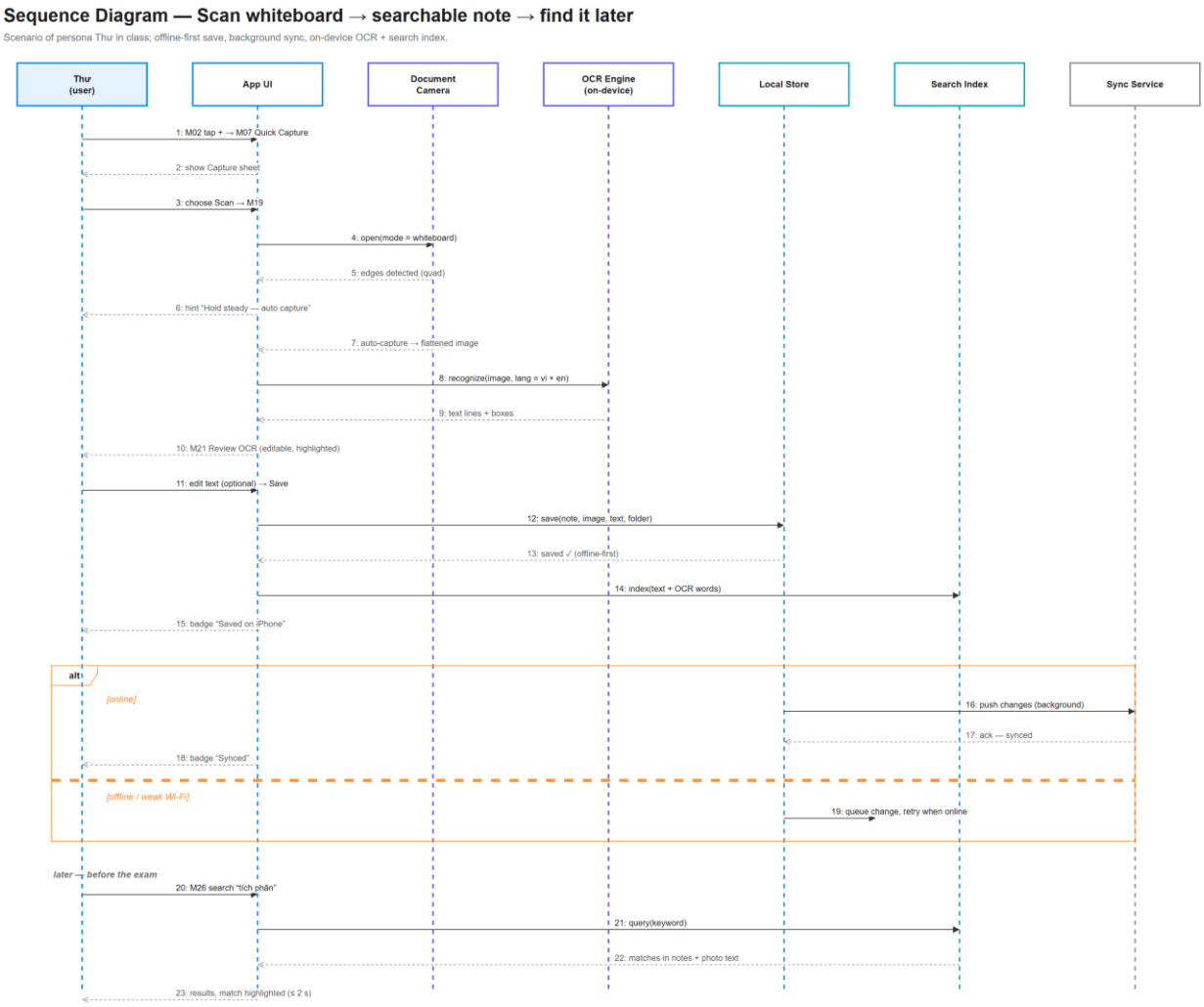


Figure 7. Sequence diagram — scan whiteboard → searchable note → find it later (with offline/online alternative)

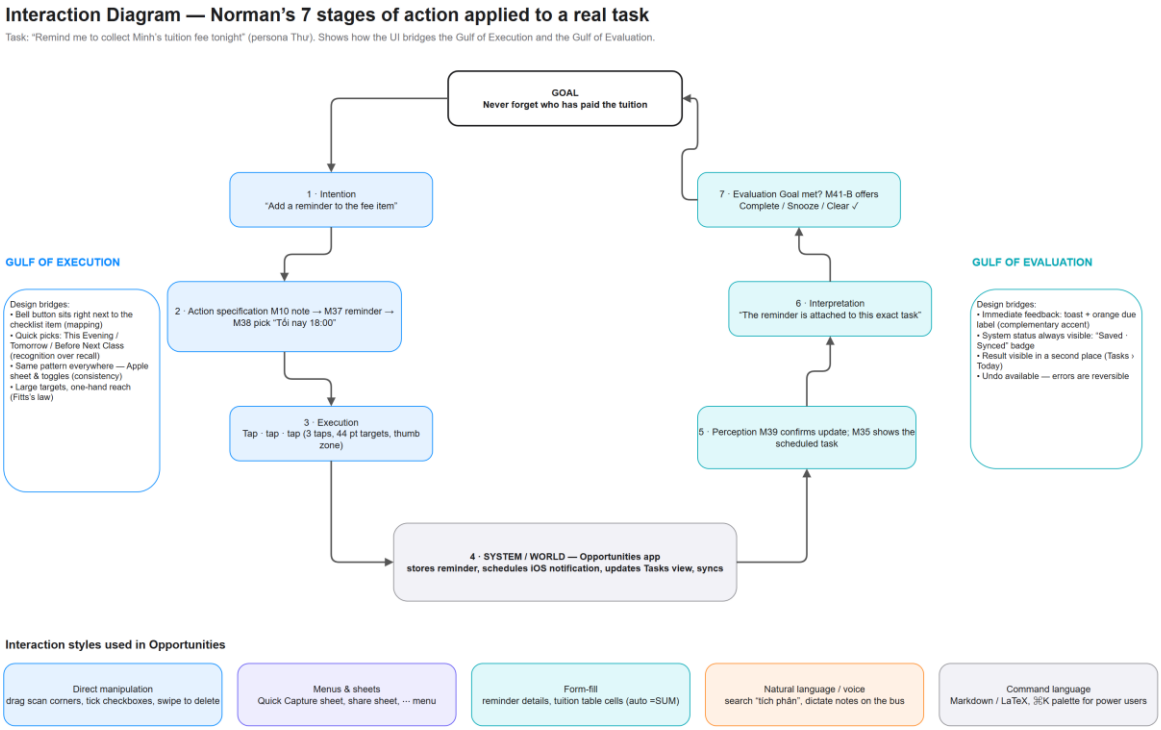


Figure 8. Interaction diagram — Norman's 7 stages of action for "remind me to collect Minh's fee tonight"

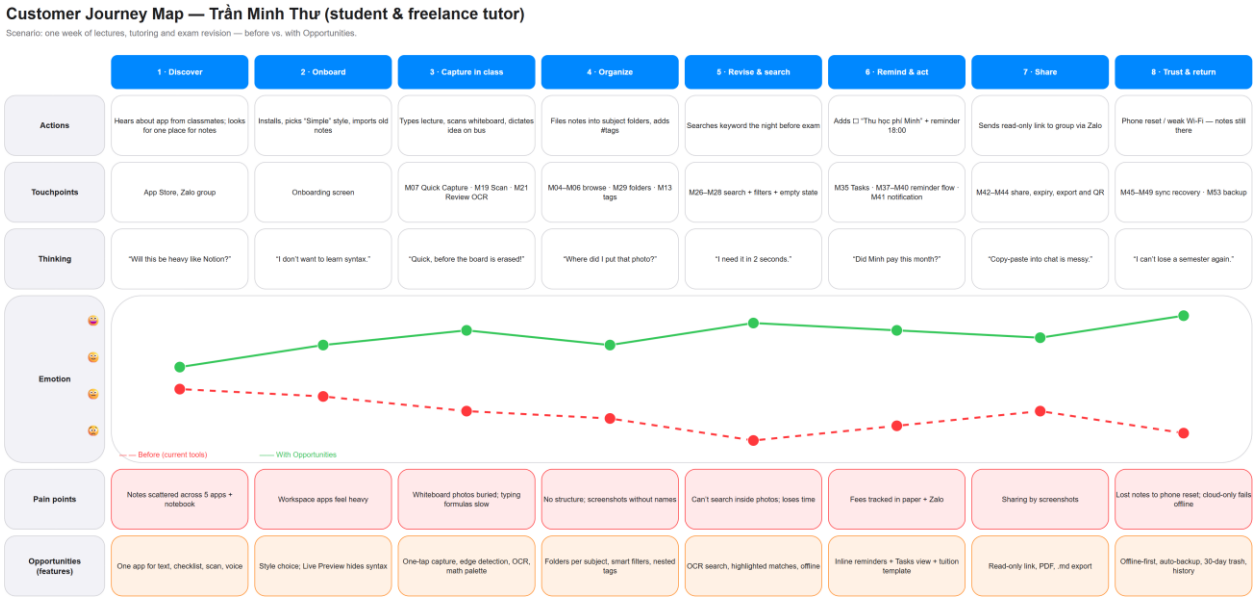


Figure 9. Customer journey map — Trần Minh Thư, before vs. with Opportunities

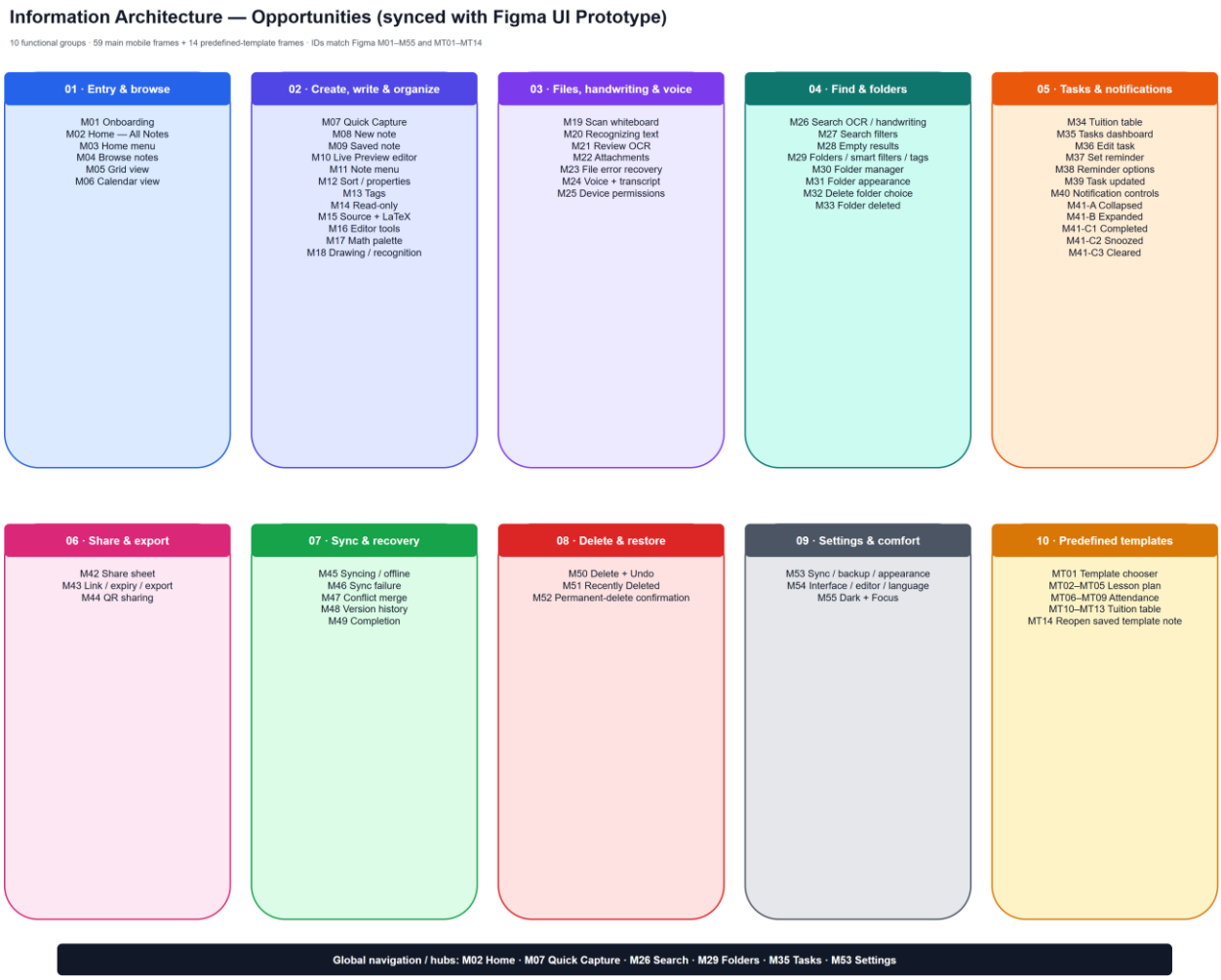


Figure 10. Information architecture — 10 functional groups covering 59 main frames and 14 predefined-template frames

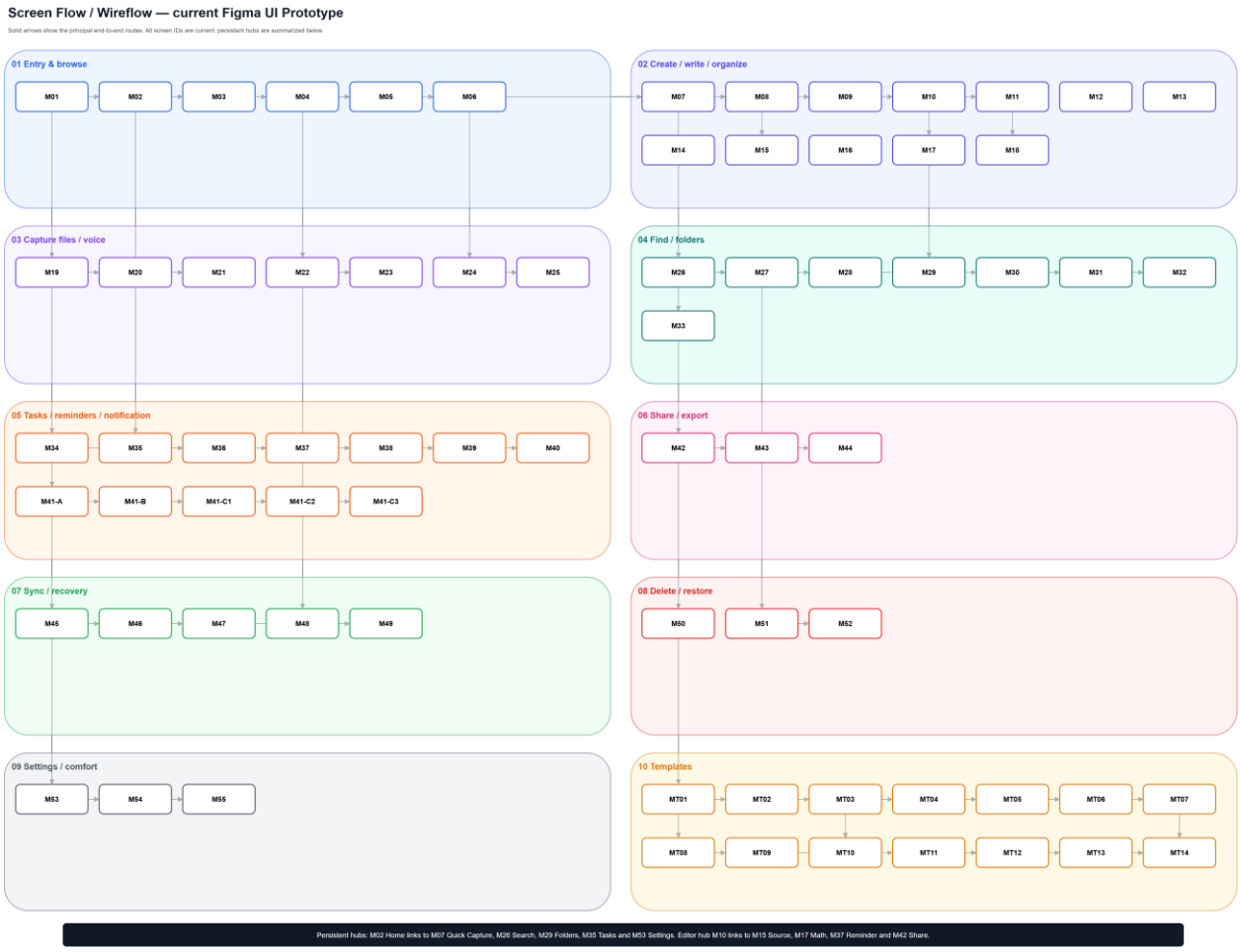


Figure 11. Screen flow / wireflow — current IDs M01–M55, M41 variants and MT01–MT14 with principal routes

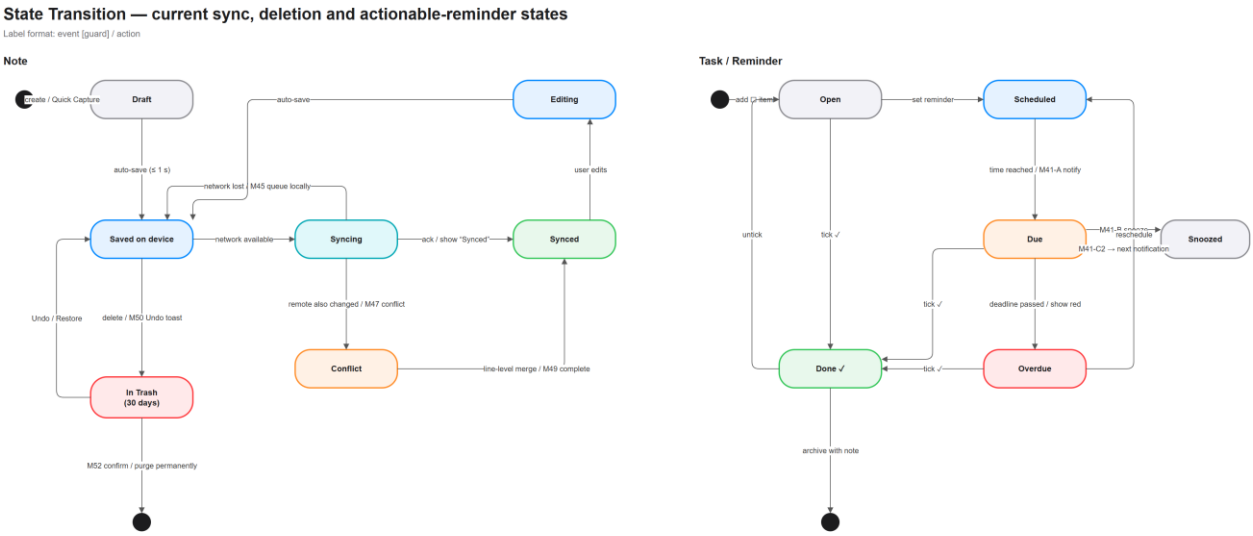


Figure 12. State transition — sync failure/conflict, deletion/restore and actionable reminder outcomes

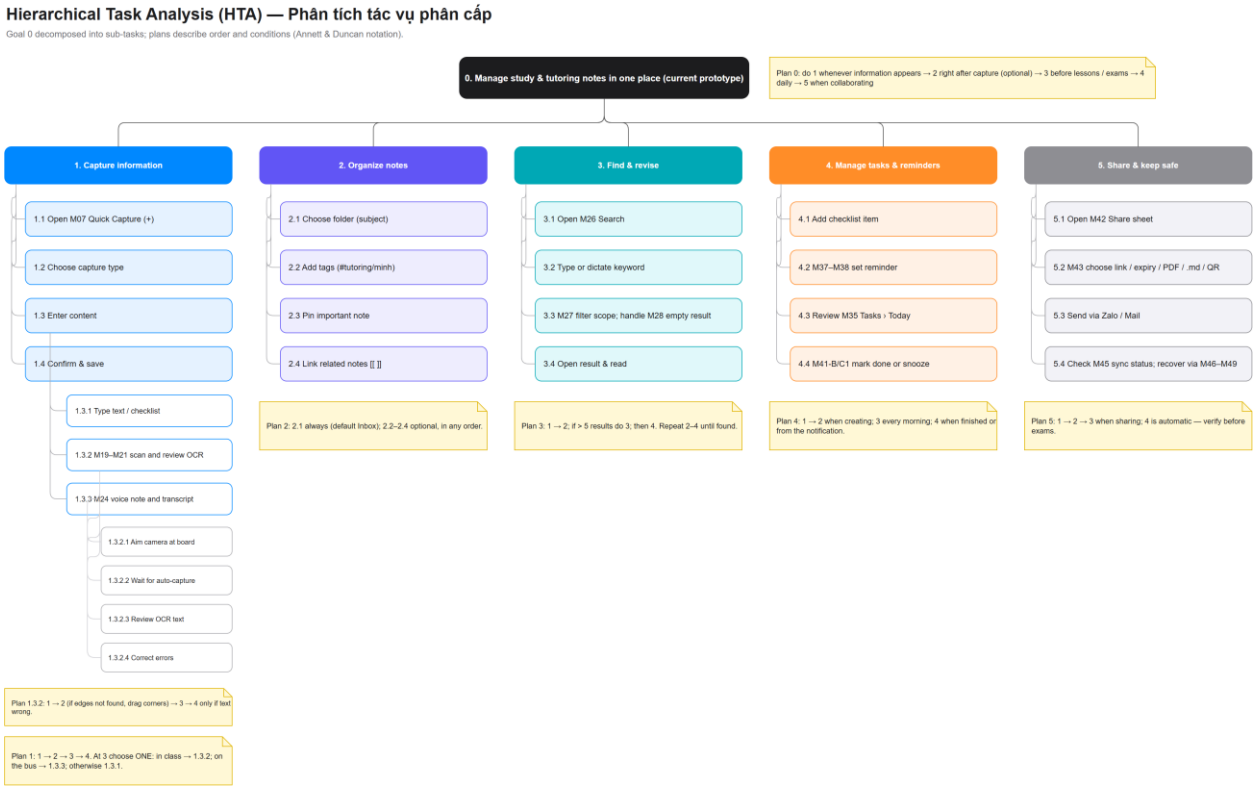


Figure 13. Hierarchical task analysis (HTA) — goal 0 “Manage study & tutoring notes in one place” with plans

B. UX wireframe

Low-fidelity wireframes use only grey blocks (bars = text, boxes = content, circles = buttons) to show structure before visual design. Each wireframe is annotated with the HCI principle applied, the design decision and the persona pain point it answers.

Mobile app wireframes (iPhone, 27 low-fidelity screens; retained as the earlier design stage)

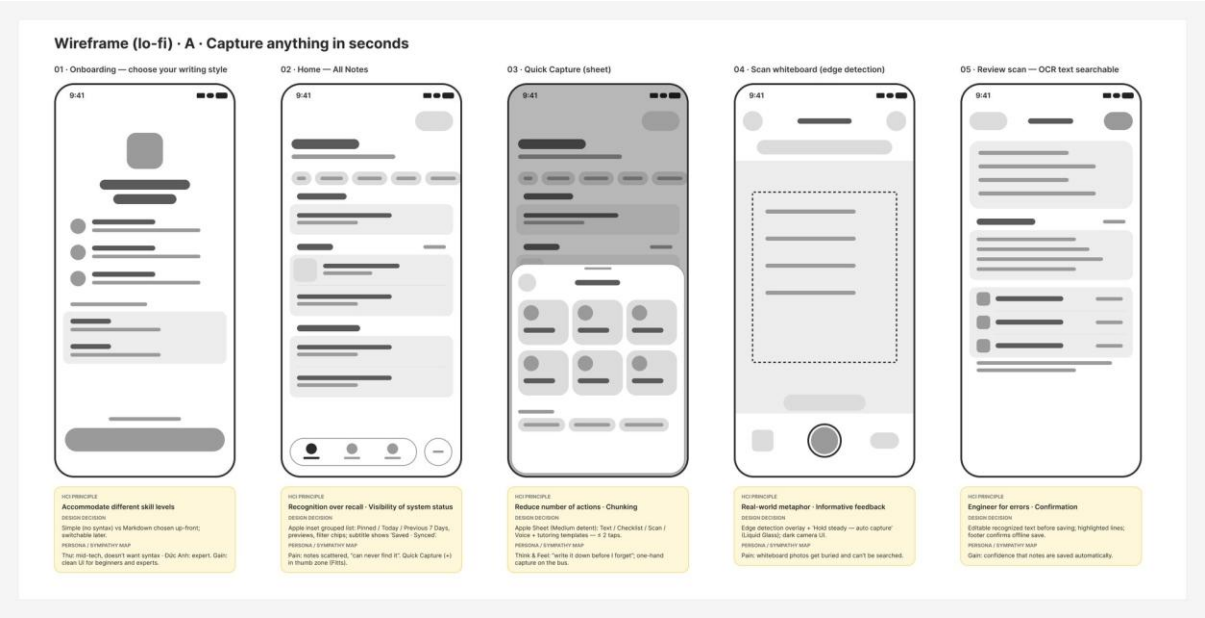


Figure 14. Mobile wireframes A — Capture anything in seconds (onboarding, home, quick capture, scan, review)

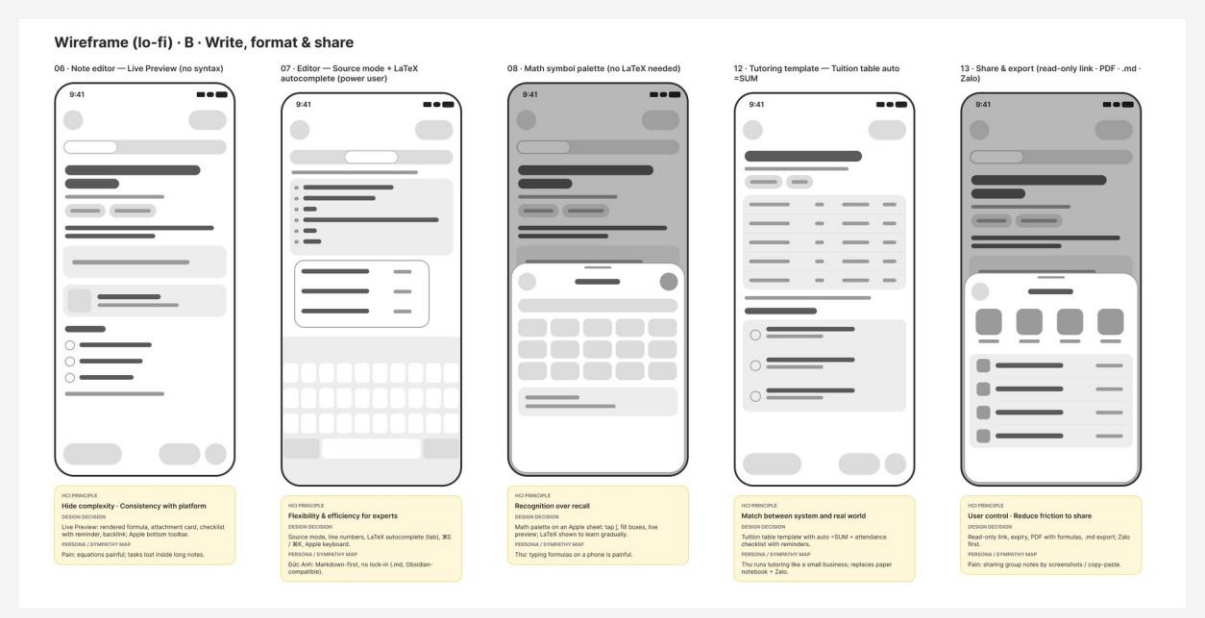


Figure 15. Mobile wireframes B — Write, format & share (editor, source mode, math palette, tuition table, share)

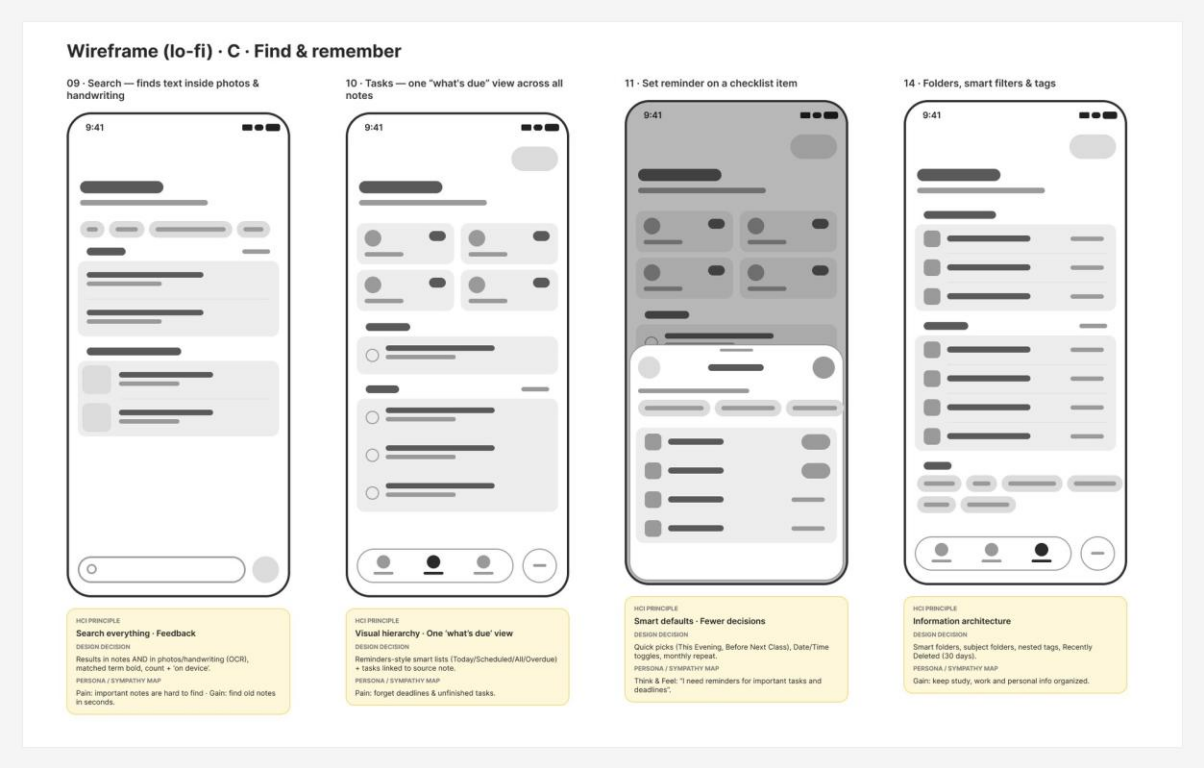


Figure 16. Mobile wireframes C — Find & remember (search, tasks, reminder, folders)

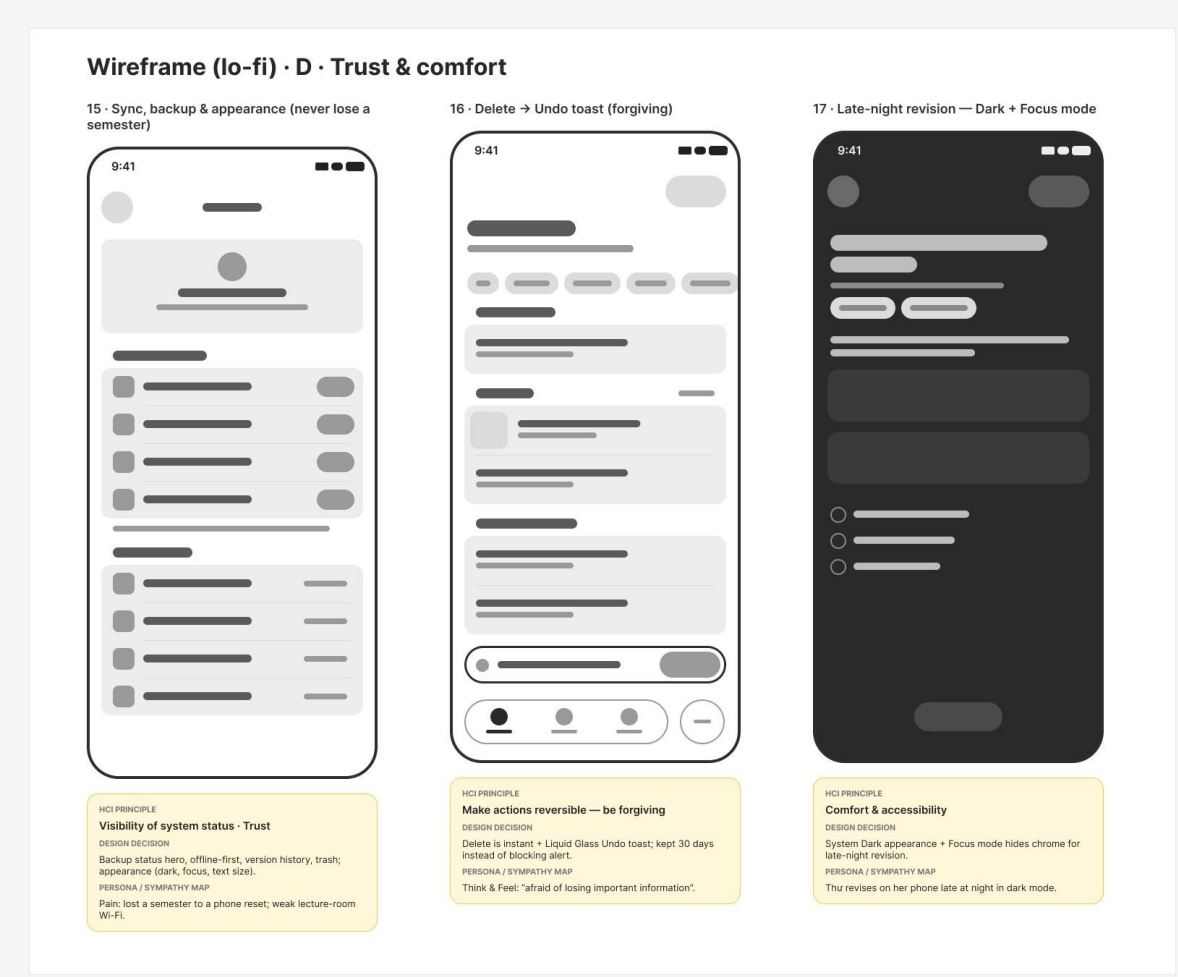


Figure 17. Mobile wireframes D — Trust & comfort (settings, undo, dark + focus)

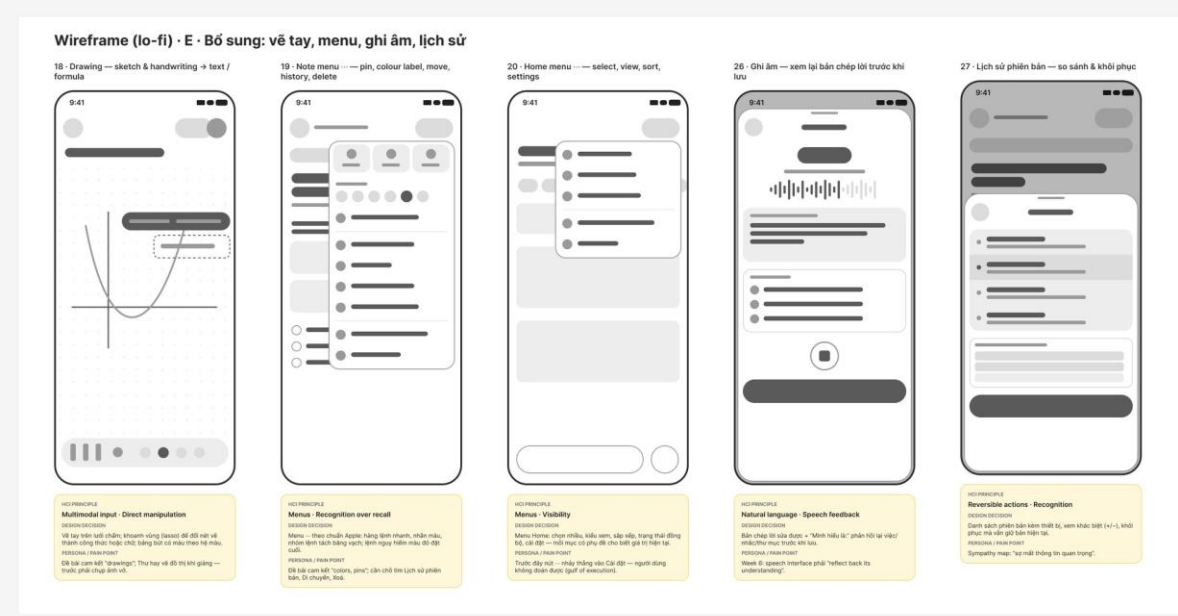


Figure 18. Mobile wireframes E — Drawing, menus, voice note, version history

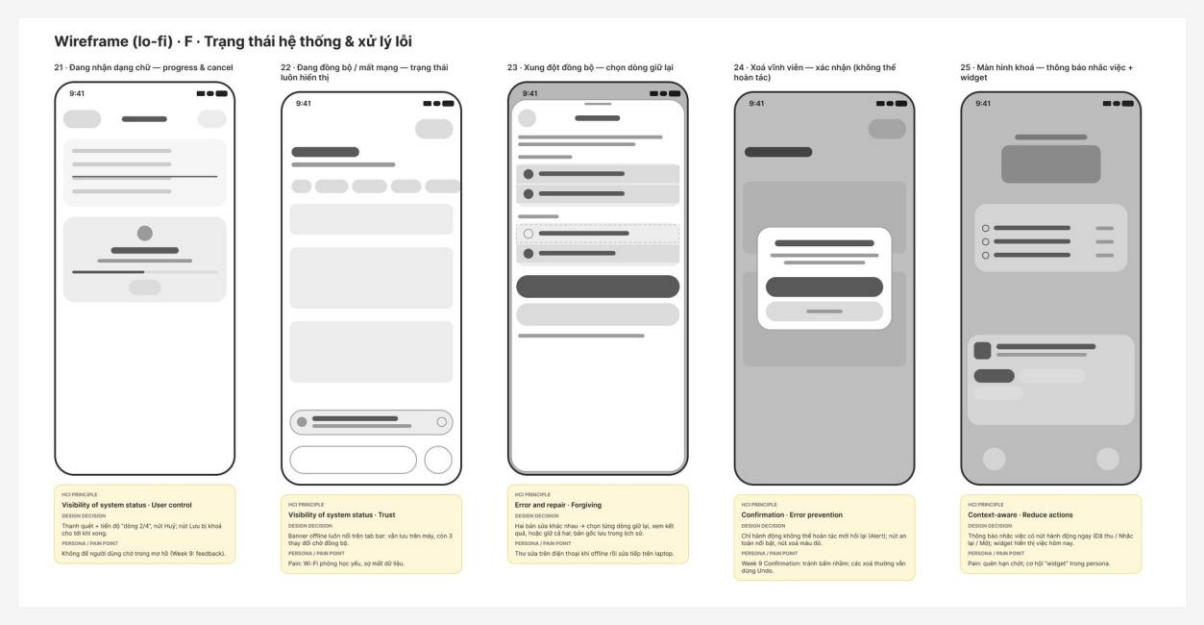


Figure 19. Mobile wireframes F — System status & error handling (processing, offline, conflict, confirmation, notification)

Web app wireframes (desktop, 12 screens)



Figure 20. Web wireframe W01 Notes — sidebar · list · editor



Figure 21. Web wireframe W02 Editor — Source | Preview split



Figure 22. Web wireframe W03 Command palette Ctrl K



Figure 23. Web wireframe W04 Search — photos & handwriting



Figure 24. Web wireframe W05 Tasks + reminder inspector



Figure 25. Web wireframe W06 Tutoring — tuition table

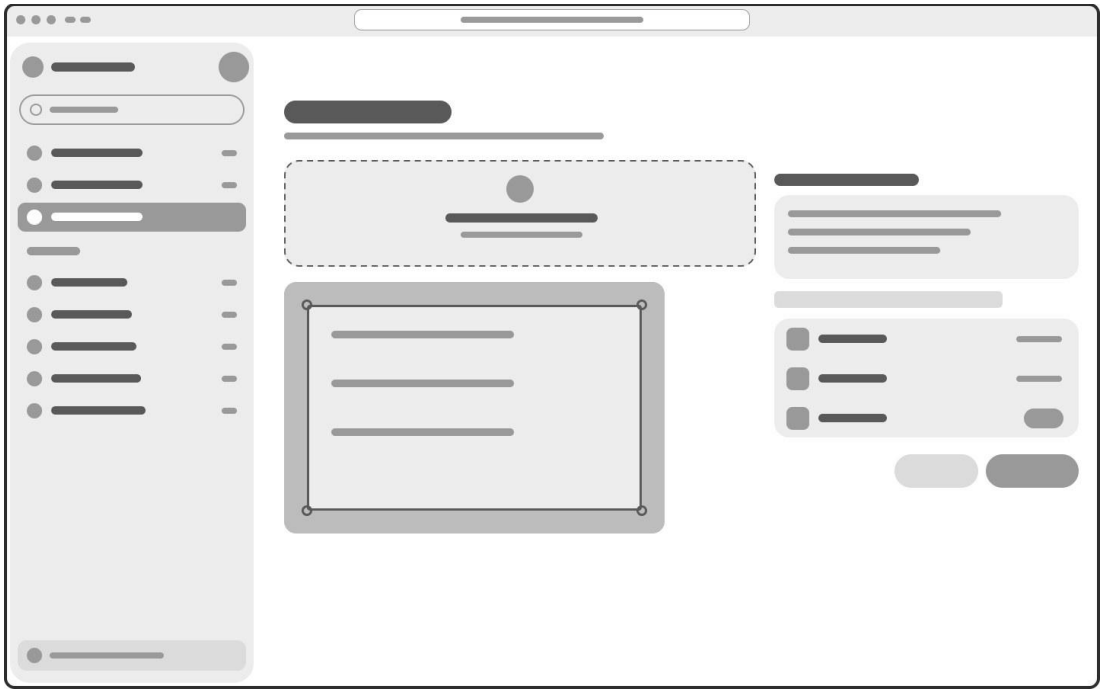


Figure 26. Web wireframe W07 Import scan — drag & drop



Figure 27. Web wireframe W08 Share & export dialog



Figure 28. Web wireframe W09 Settings — sync & backup



Figure 29. Web wireframe W10 Dark + Focus mode



Figure 30. Web wireframe W11 Right-click menu + tooltips



Figure 31. Web wireframe W12 Sync conflict — merge line by line

IV. Test cases

Functional test cases verify that each key flow of the prototype works as specified; the usability test plan below measures whether real users can complete the persona tasks efficiently.

ID	Test case	Precondition	Test steps	Expected result	Screens
TC-01	Quick capture a text note	App installed; user on Home	1. Tap “+” 2. Choose Văn bản (Text) 3. Type a title and content 4. Go back	Note appears at the top of “Hôm nay” with badge “Đã lưu”; total taps to start writing ≤ 2	M02 → M07 → M08 → M09
TC-02	Scan a whiteboard and make it searchable	Camera permission granted	1. Tap “+” → Quét ảnh 2. Point at the board until edges are detected 3. Wait for auto-capture 4. Review recognized text, edit one word 5. Choose folder, tap Lưu	Progress “Đang nhận dạng chữ…” shown, Save disabled until done; note saved with photo + editable text	M07 → M19 → M20 → M21 → M09
TC-03	Find text inside a photo	TC-02 completed	1. Tap Search 2. Type “tích phân” 3. Filter “Ảnh & chữ viết tay” 4. Open the first result	Results within 2 s, match highlighted in the photo; works with Wi-Fi off	M26 → M27 → M21 / M10; W04
TC-04	Insert a formula without LaTeX	Note open in Live Preview	1. Tap Σ on the toolbar 2. Tap ∫, fill limits 0 and 2π 3. Tap Insert	Rendered formula appears in the note; LaTeX source visible only in Source mode	M10 → M17 → M10
TC-05	Set a reminder on a checklist item	Note contains an unchecked item	1. Tap the bell next to the item 2. Pick “Tối nay 18:00” 3. Tap ✓	Item shows orange due label; appears in Tasks › Hôm nay; notification fires at 18:00 with “Đã thu ✓” action	M10 → M37 → M39 → M35 → M41-B/C1
TC-06	Work offline and sync later	Airplane mode on	1. Edit a note 2. Observe the banner 3. Turn Wi-Fi on	Banner “Đang offline — vẫn lưu trên máy”; changes sync automatically when online; badge “Đã đồng bộ”	M45 → M46 → M02

ID	Test case	Precondition	Test steps	Expected result	Screens
TC-07	Resolve a sync conflict	Same note edited on phone (offline) and web	1. Reconnect 2. In the merge dialog, select lines to keep 3. Tap “Gộp các dòng đã chọn”	Merged note contains all selected lines; both originals kept in version history	M47 → M49; W12
TC-08	Delete and undo	At least one note exists	1. Swipe a note → Delete 2. Tap Hoàn tác in the toast	Note restored to the same position; without undo it stays 30 days in Recently Deleted	M50 → M02
TC-09	Permanently delete with confirmation	Notes in Recently Deleted	1. Open Recently Deleted 2. Select 3 notes → Delete permanently	Alert “Xoá vĩnh viễn 3 ghi chú?” with safe default “Giữ lại”; notes removed only after confirming	M51 → M52
TC-10	Colour label and pin from the ··· menu	Note open	1. Tap ··· 2. Tap Ghim and choose the orange label	Note pinned at top of Home with orange dot; menu shows grouped actions and destructive action last	M10 → M11 → M12
TC-11	Share a read-only link	Note open	1. Tap Share 2. Toggle Link chỉ xem, set expiry 3. Tap Sao chép link	Link copied; recipient can view in a browser without an account	M42 → M43 → M44; W08
TC-12	Keyboard-first editing on web	Web app open on Windows	1. Press Ctrl K, type “ins”, Enter on Insert formula 2. Press Ctrl E to switch to Source 3. Right-click a note in the list	Palette filters commands with shortcuts; mode switches; context menu shows Ctrl shortcuts	W03 · W02 · W11
TC-13	Create and reopen a note from a predefined template	User on Home; no template draft open	1. Tap + → Mẫu 2. Choose Giáo án, Điểm danh or Bảng học phí 3. Preview and tap Dùng mẫu này 4. Edit a field and save 5. Open Ghi chú từ mẫu	A separate note is saved without changing the original template; edited values persist and the note can be reopened.	M07 → MT01 → MT02/MT06/MT10 → MT03/MT07/MT11 → MT04/MT08/MT12 → MT05/MT09/MT13 → MT14

Usability test plan

Task given to participant	Persona	Success criterion	Metric
“Save this whiteboard photo so you can find it later”	Thư	Note saved with recognized text	≤ 4 taps · ≤ 20 s
“Find the note that mentions ‘tích phân’ in a photo”	Both	Correct result opened	≤ 3 actions · ≤ 10 s
“Remind yourself to collect Minh’s fee tonight”	Thư	Reminder set on checklist item	≤ 4 taps
“Insert ∫ from 0 to 2π of f(θ)dθ without typing code”	Thư	Formula inserted correctly	0 errors
“Write the same formula in Source mode and export .md”	Đức Anh	Valid .md with LaTeX	≤ 60 s
“You deleted a note by mistake — get it back”	Both	Note restored	≤ 2 taps

Method: 5 participants per persona, think-aloud protocol, System Usability Scale (SUS) questionnaire after the tasks; any task under 80% success is redesigned in the next iteration.

Heuristic evaluation (v1 → v2)

Heuristic	Problem found in v1	Severity	Fix in v2
Visibility of system status	No processing / offline / syncing states	3	M20 recognition progress; M45 syncing/offline; M46 retry/continue offline
Match with real world	Jargon (OCR, =SUM, .md) in Simple mode; mixed languages	3	Vietnamese, jargon-free copy; M54 language and editor preferences
User control & freedom	··· jumped straight to Settings; no Cancel while processing	3	Full menus M03/M11/W11; Cancel on M20; explicit back/cancel paths
Consistency & standards	⌘ shown to Windows users; two icon styles	2	Ctrl/⌘ by OS; one icon set; consistent list, sheet and action styles
Error prevention	Permanent delete without confirmation	4	M32 folder-deletion choice and M52 permanent-delete confirmation
Recognition rather than recall	Icon-only buttons without labels	2	Labelled menus, tooltips and M16 editor-tools catalogue
Help users recover from errors	No UI for sync conflicts	4	M23 file recovery; M46–M49 sync recovery; M48 version history; W12 merge

PART 2 – PROTOTYPE

The hi-fi prototype is built in Figma on top of the official Apple iOS & iPadOS 27 library (real components: status bar, toolbars, Liquid Glass buttons, tab bar, lists, sheets, alerts, keyboard), with a colour system defined on the colour wheel and a component state matrix.

Figma design file:
https://www.figma.com/design/VWLZhe9YBBkKE3Fl8mNfH5/Oppotunities

Visual design system

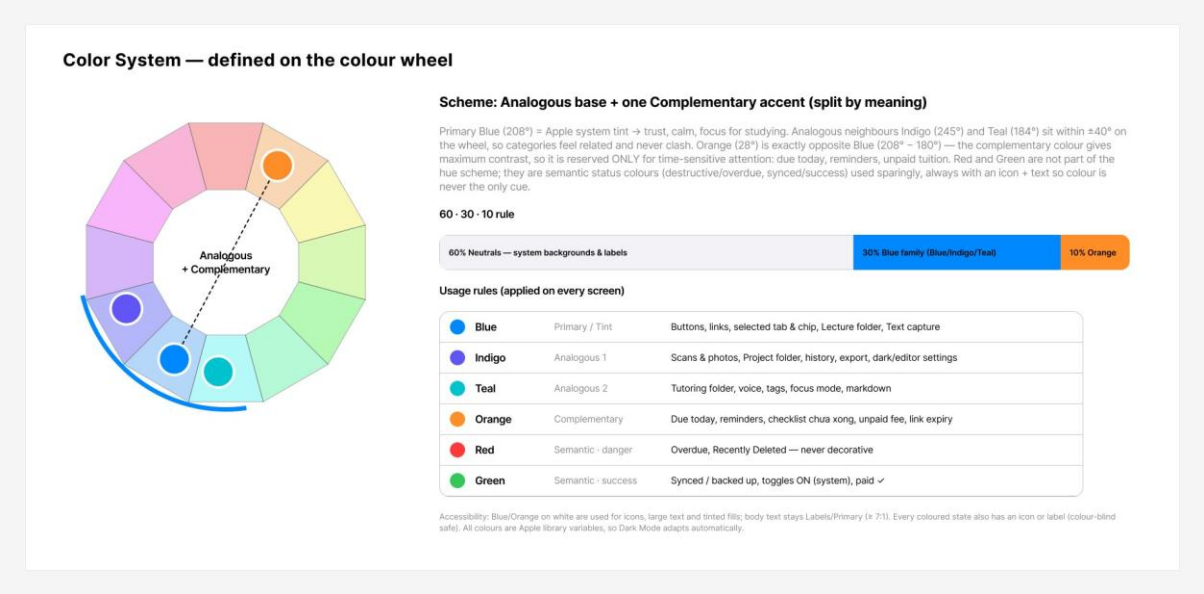


Figure 32. Colour system — Blue primary (208°), analogous Indigo/Teal, complementary Orange (28°), 60-30-10 rule

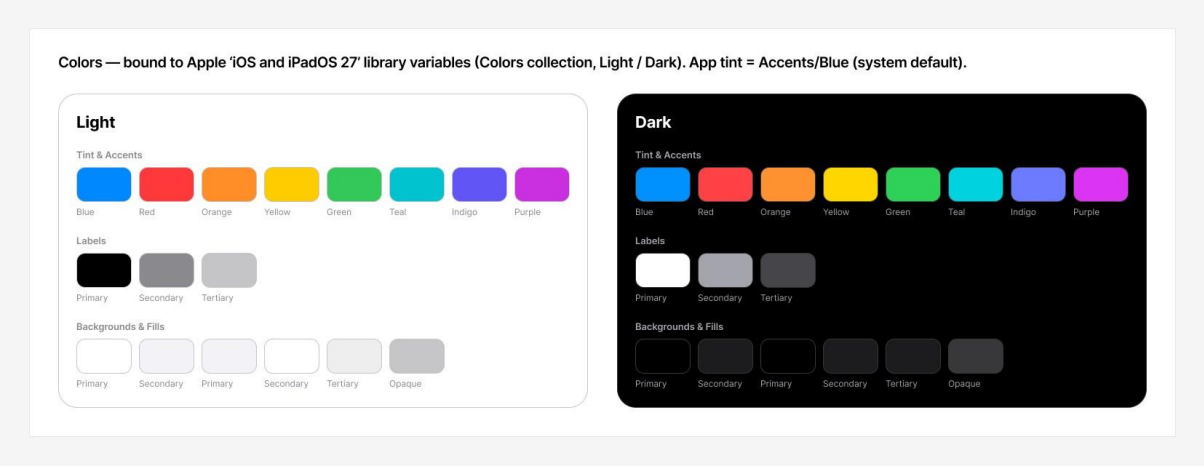


Figure 33. Apple iOS 27 system colour variables, Light and Dark modes

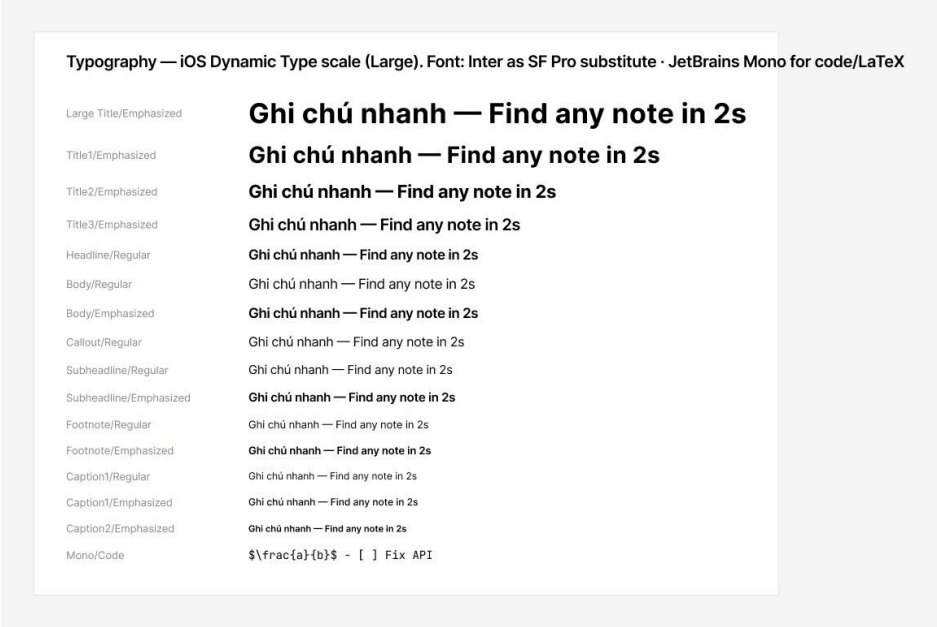


Figure 34. Typography — Dynamic Type scale (Large default)

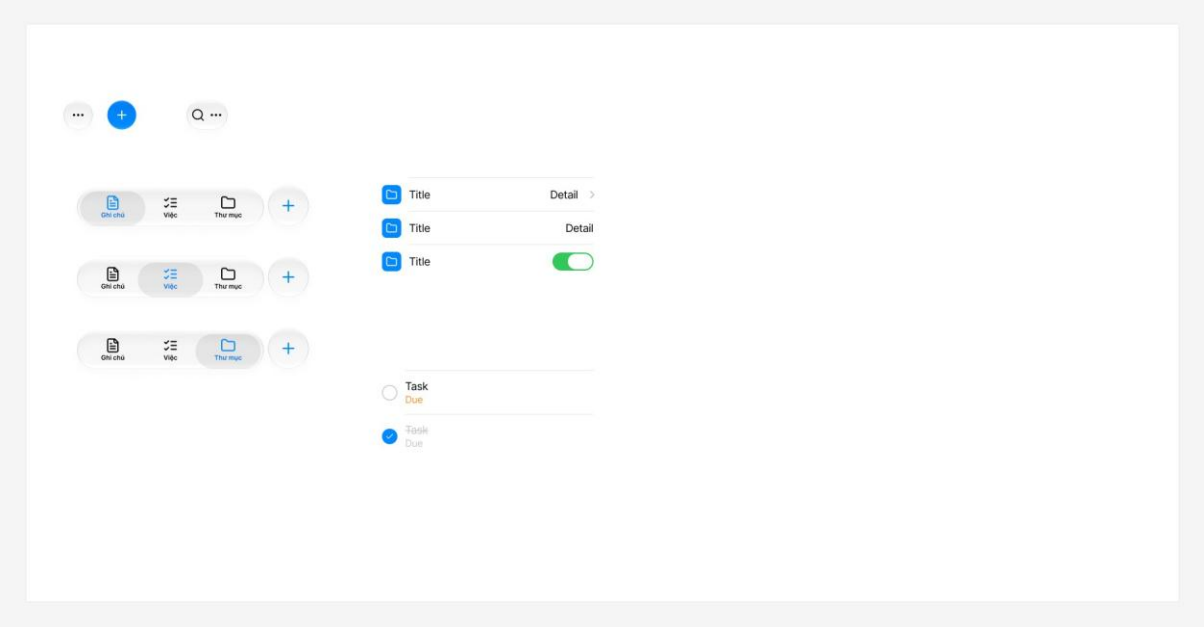


Figure 35. Components derived from the Apple kit — glass buttons, tab bar, list rows, task rows

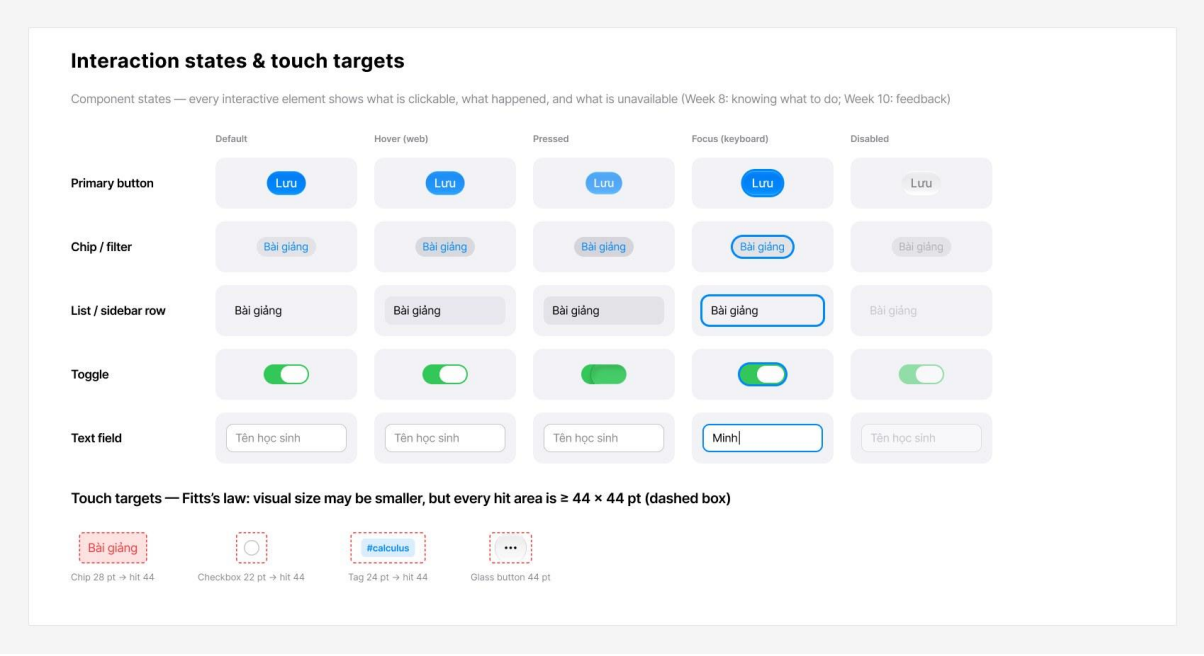


Figure 36. Interaction states (default, hover, pressed, focus, disabled) and 44 pt touch targets

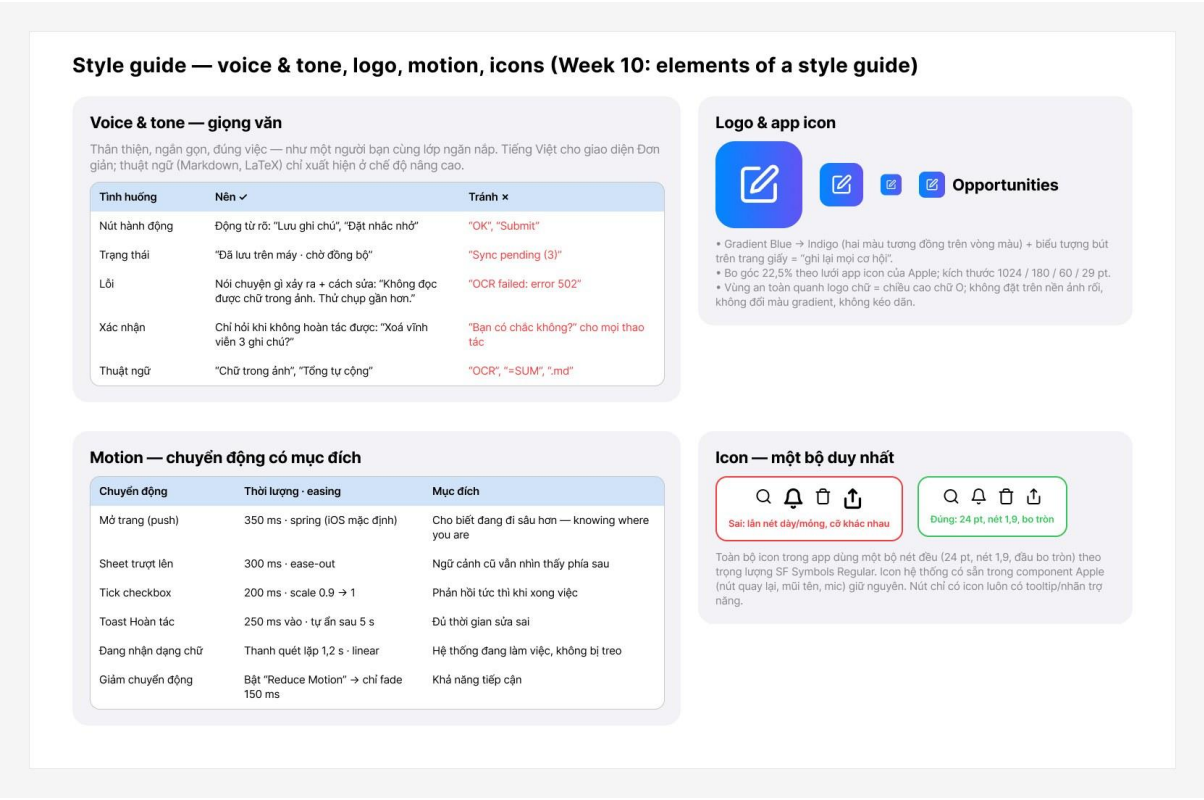


Figure 37. Style guide — voice & tone, logo/app icon, motion, icon rules

I. Mobile app

Mobile prototype (Figma, start at 01 Onboarding):
https://www.figma.com/proto/VWLZhe9YBBkKE3Fl8mNfH5/Oppotunities?page-id=6%3A5&node-id=41-2326&starting-point-node-id=41%3A2326&scaling=scale-down

The current mobile prototype contains 59 main iPhone frames across nine functional flows, plus 14 predefined-template frames in a tenth flow (73 UI frames total). Hotspots connect the principal scenarios end to end, including browse modes, error recovery, actionable notifications, QR sharing, sync conflict recovery and template reuse.

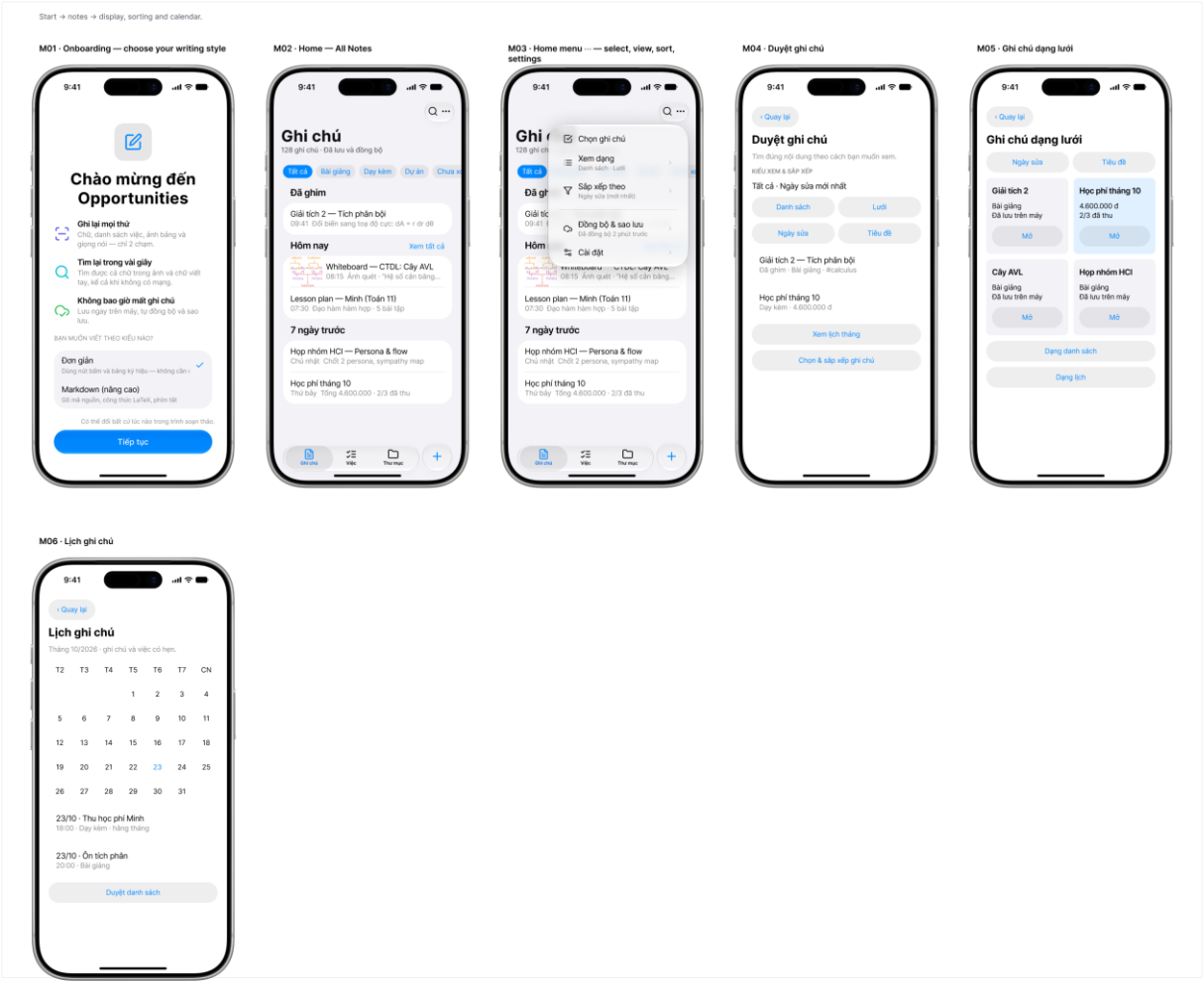


Figure 38. Mobile UI 01 — Entry & browse: M01–M06 onboarding, Home, menu, list, grid and calendar

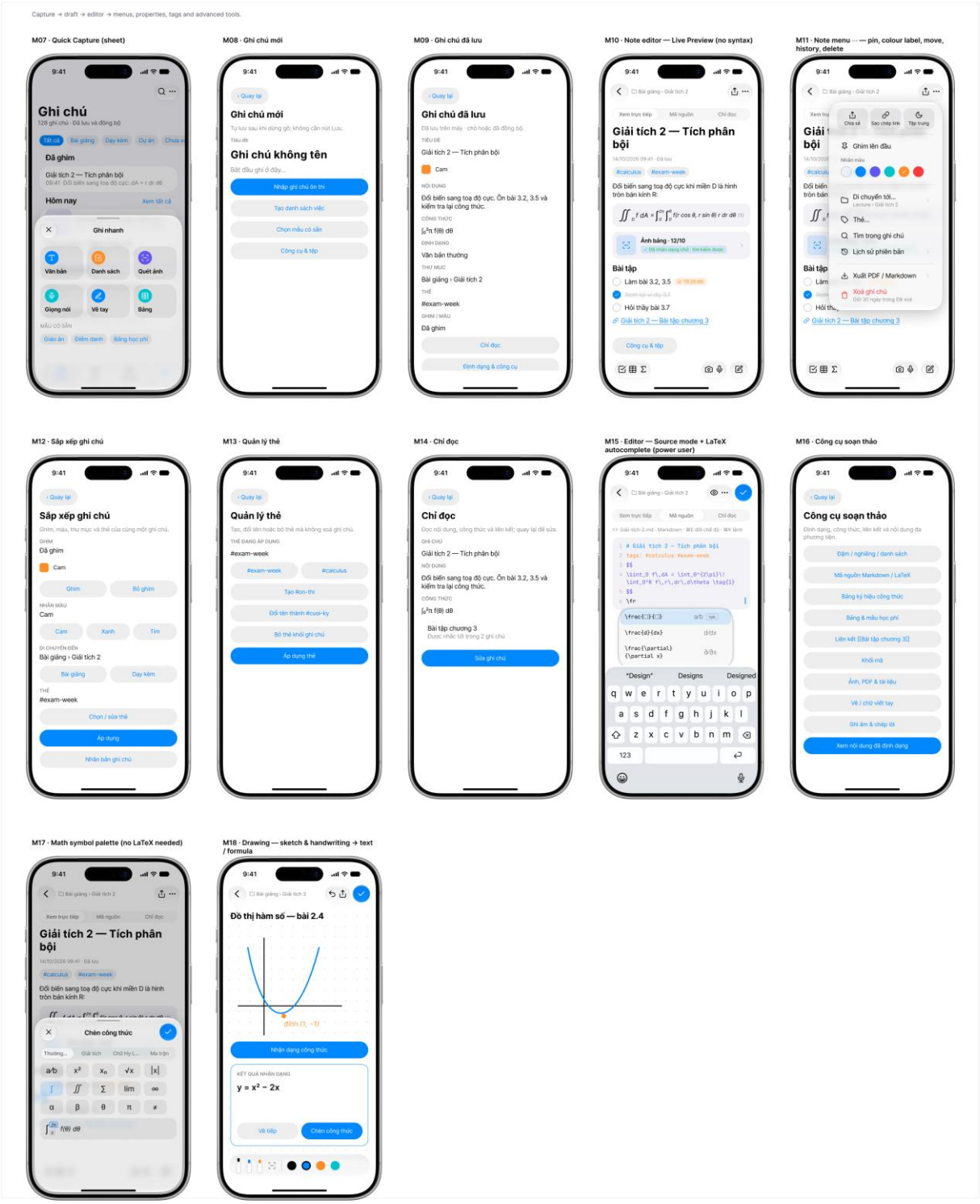


Figure 39. Mobile UI 02 — Create, write & organize: M07–M18 capture, note states, editor tools, Source, math and drawing

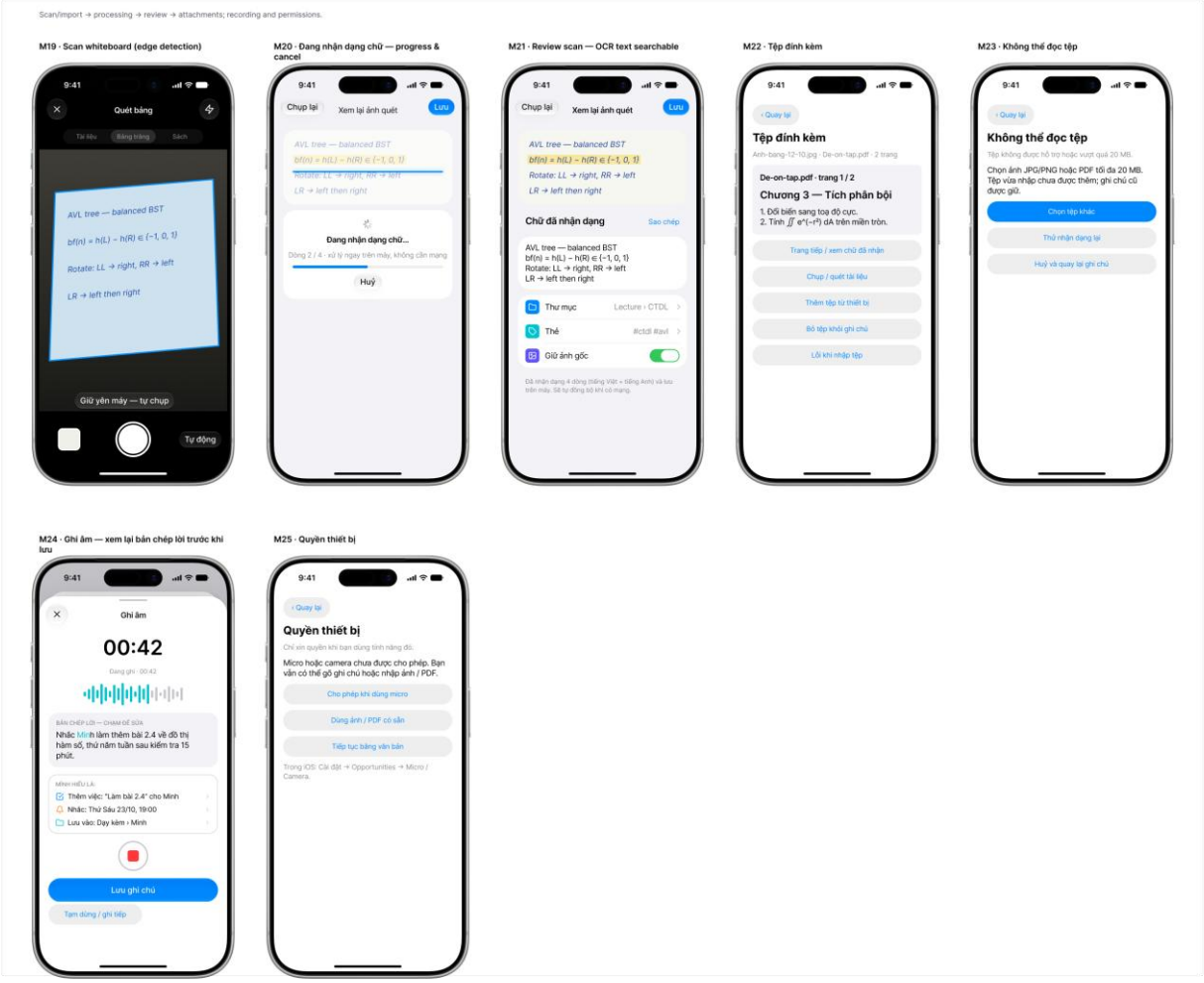


Figure 40. Mobile UI 03 — Files, handwriting & voice: M19–M25 scan, OCR, attachments, errors, transcript and permissions

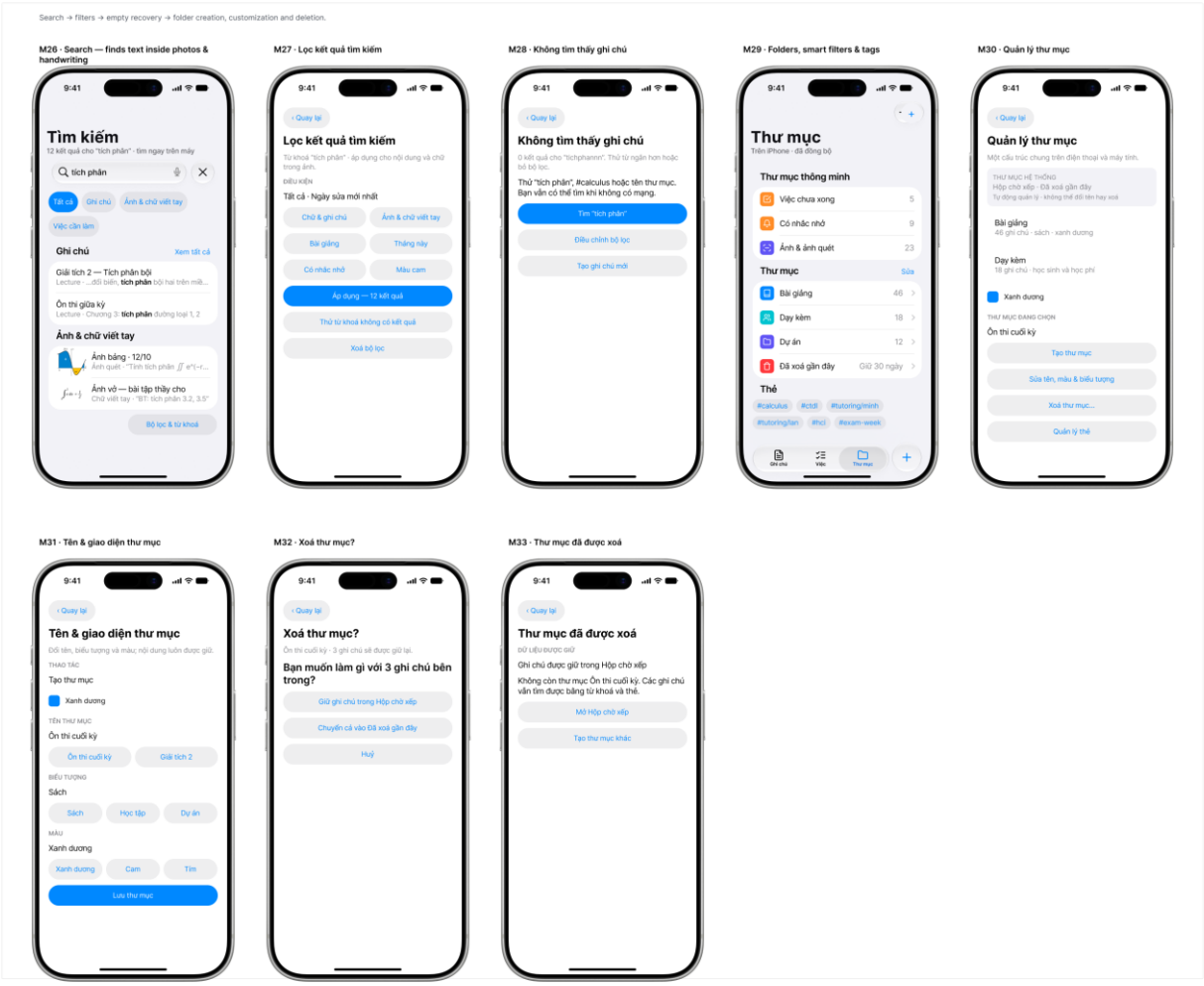


Figure 41. Mobile UI 04 — Find & folders: M26–M33 OCR search, filters, empty state and folder lifecycle

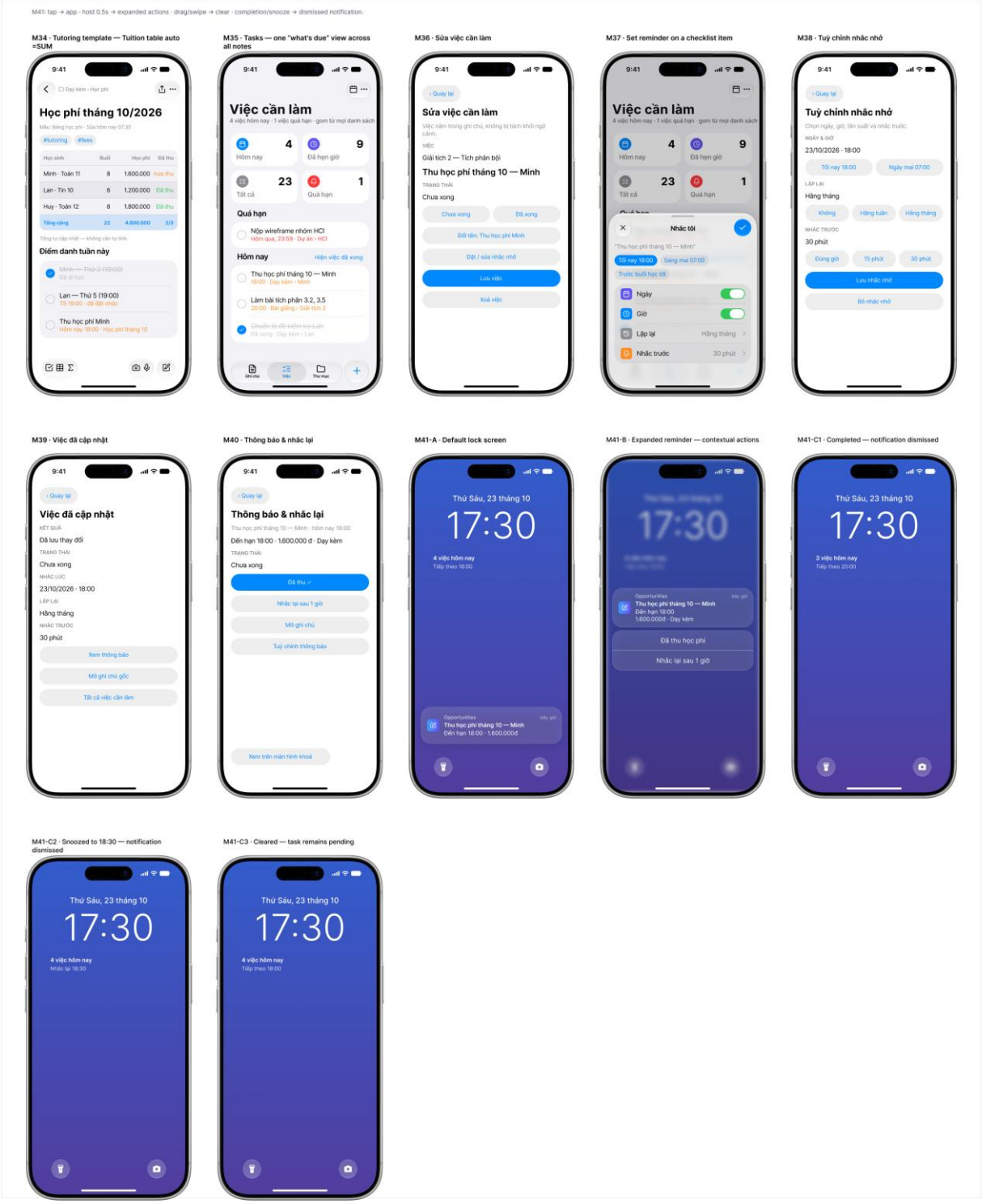


Figure 42. Mobile UI 05 — Tasks, reminders & notifications: M34–M41-C3 from tuition table to complete, snooze or clear

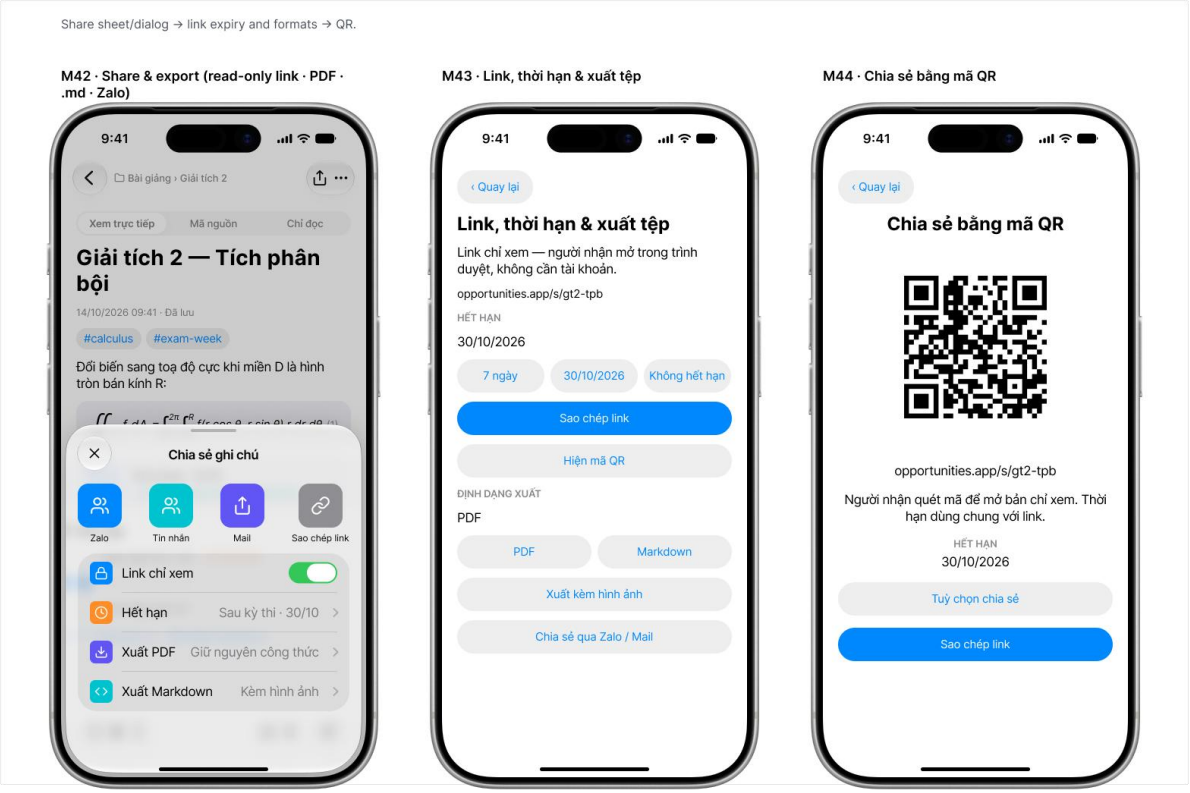


Figure 43. Mobile UI 06 — Share & export: M42–M44 read-only link, expiry, PDF/Markdown and QR

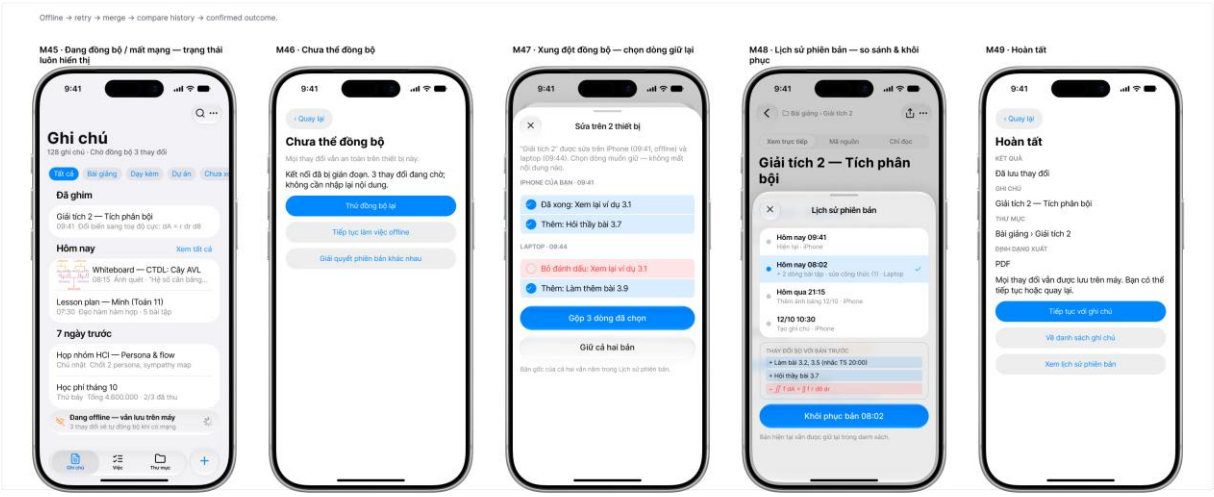


Figure 44. Mobile UI 07 — Sync & recovery: M45–M49 offline status, failure, conflict merge, version history and completion

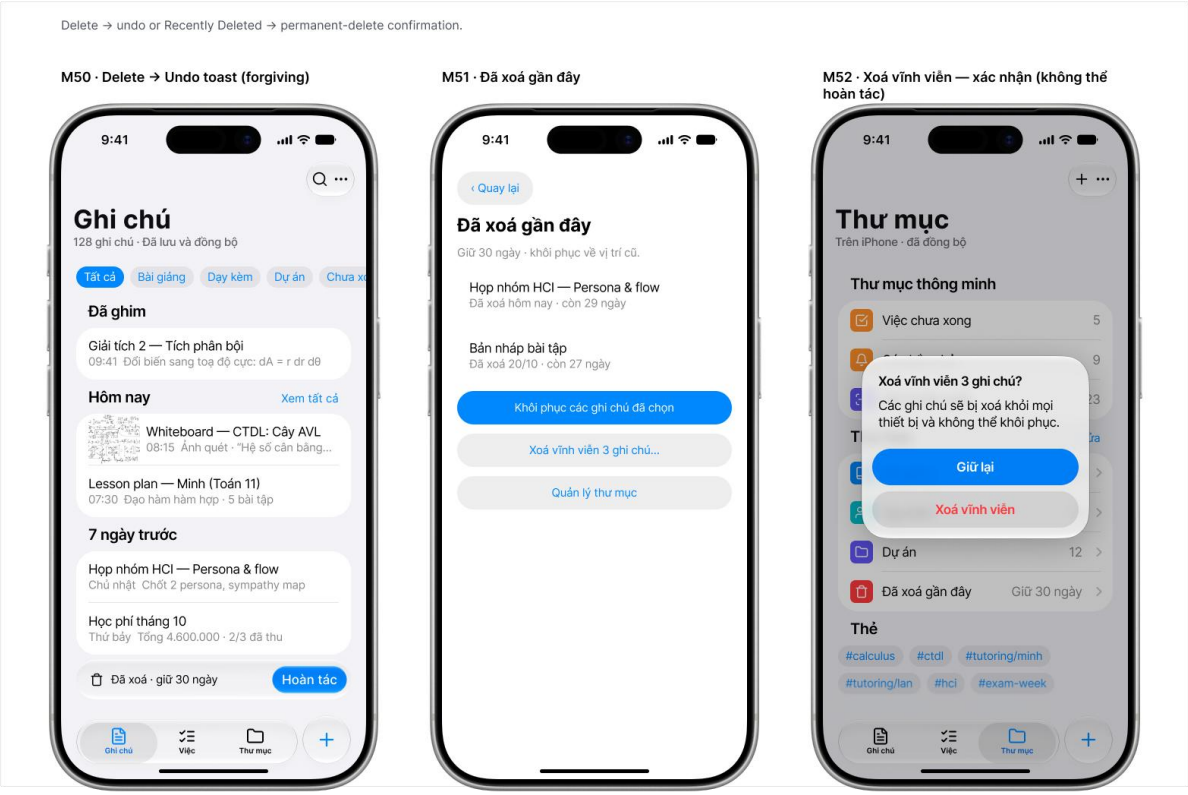


Figure 45. Mobile UI 08 — Delete & restore: M50–M52 Undo, Recently Deleted and permanent-delete confirmation

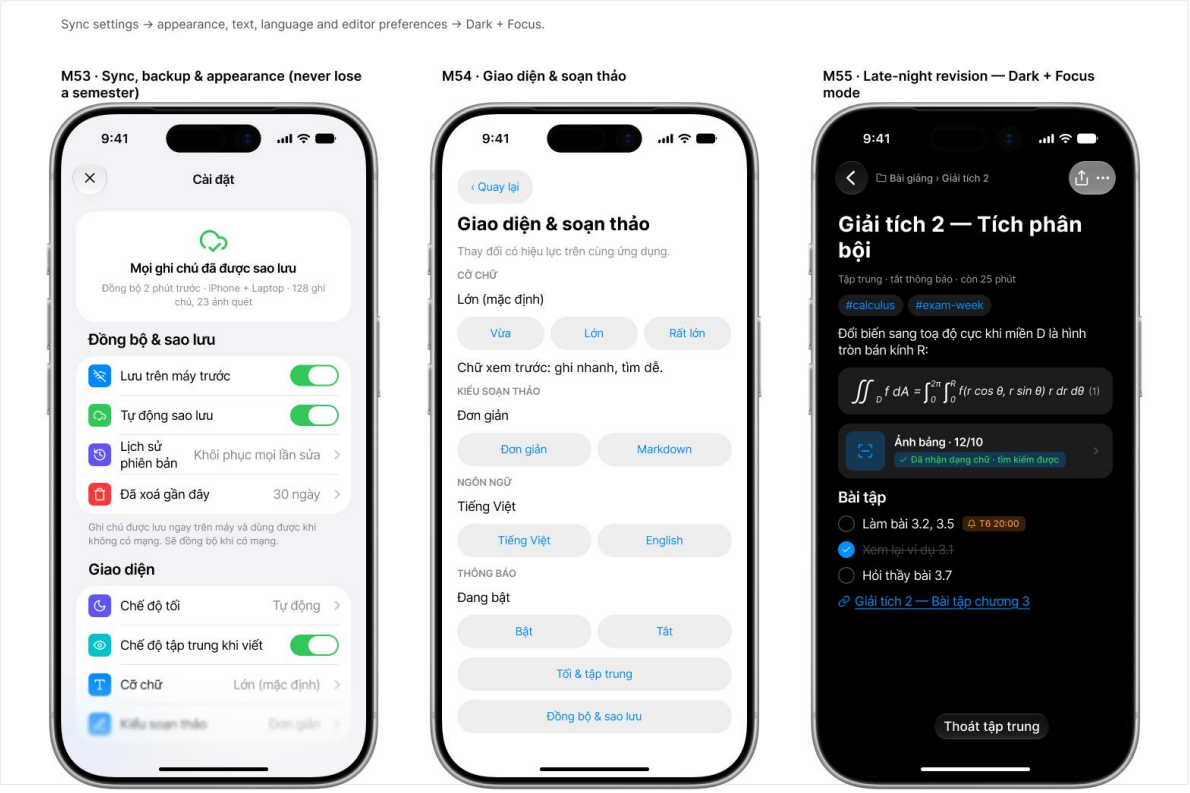


Figure 46. Mobile UI 09 — Settings & comfort: M53–M55 backup, language/editor preferences, Dark and Focus

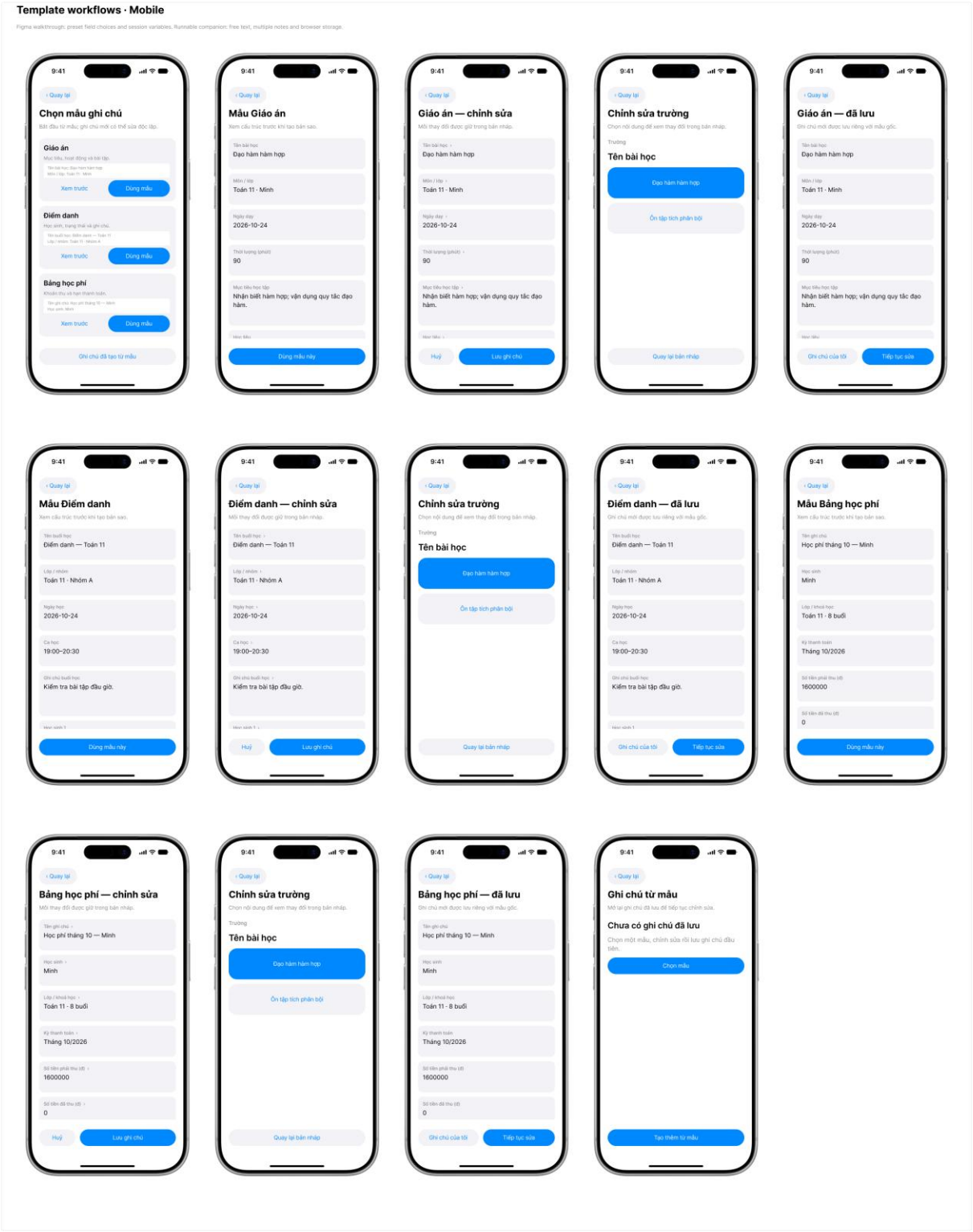


Figure 47. Mobile UI 10 — Predefined templates: MT01–MT14 browse, preview, edit, save and reopen lesson, attendance and tuition notes

II. Web

Web app prototype (Figma, start at W01):
https://www.figma.com/proto/VWLZhe9YBBkKE3Fl8mNfH5/Oppotunities?page-id=64%3A3&node-id=64-6&starting-point-node-id=64%3A6&scaling=scale-down

12 desktop screens (1440 × 900) using the iPad sidebar and toolbar components: a 3-column workspace for Thư on her Windows laptop and keyboard-first tools for Đức Anh.

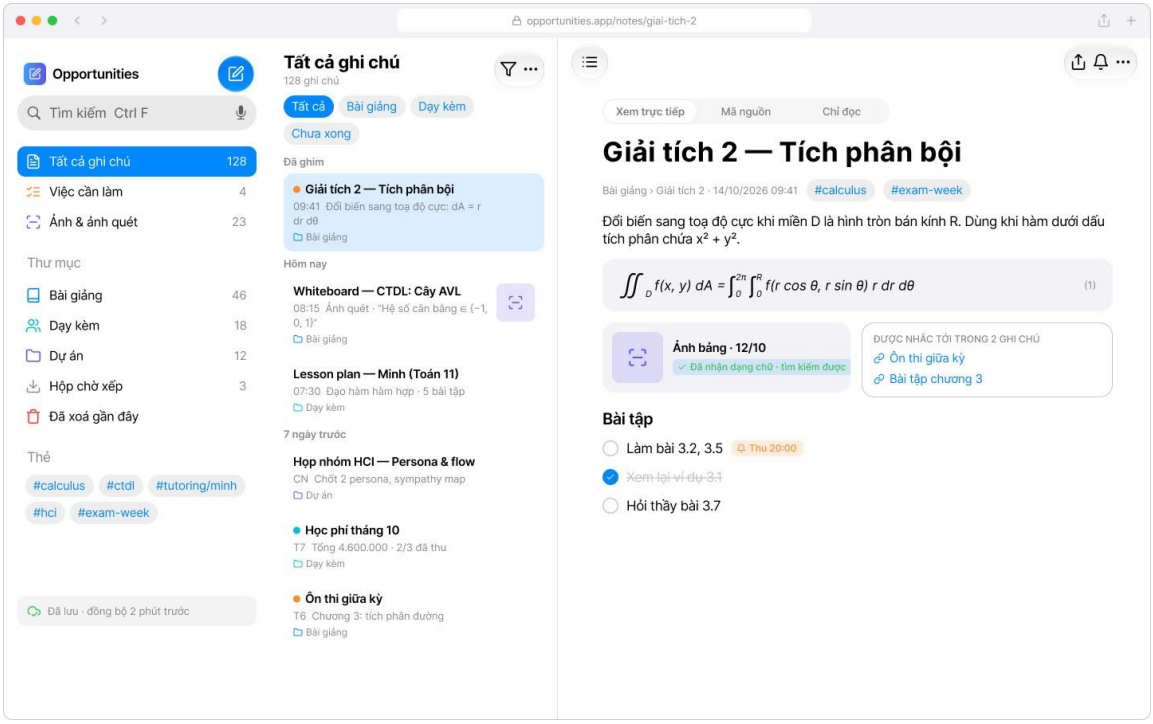


Figure 48. W01 — Notes — sidebar, note list with colour labels and editor in Live Preview

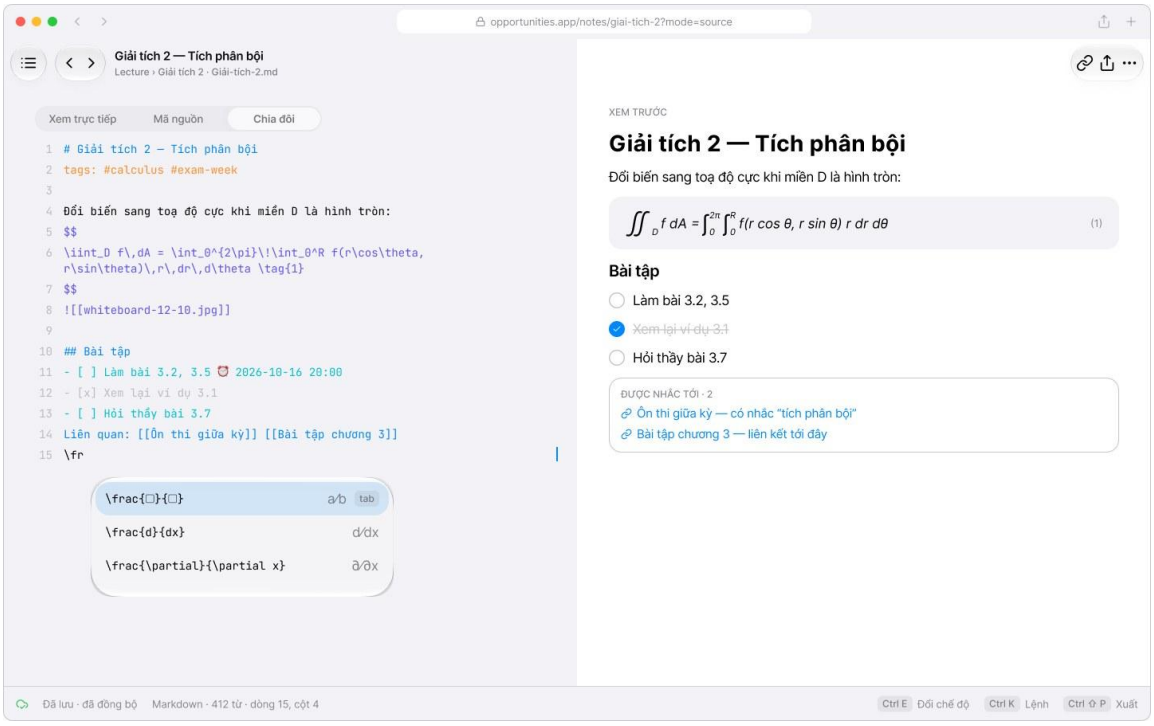


Figure 49. W02 — Editor — Source | Preview split with LaTeX autocomplete and shortcut status bar

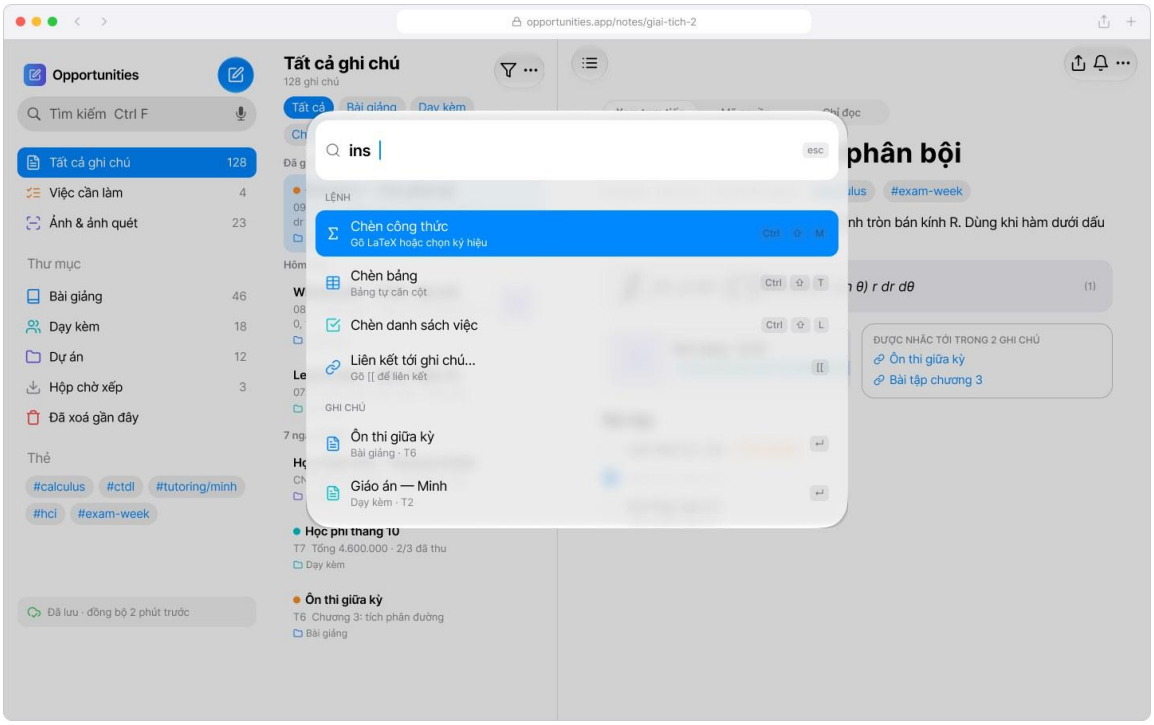


Figure 50. W03 — Command palette (Ctrl K) listing commands with their shortcuts

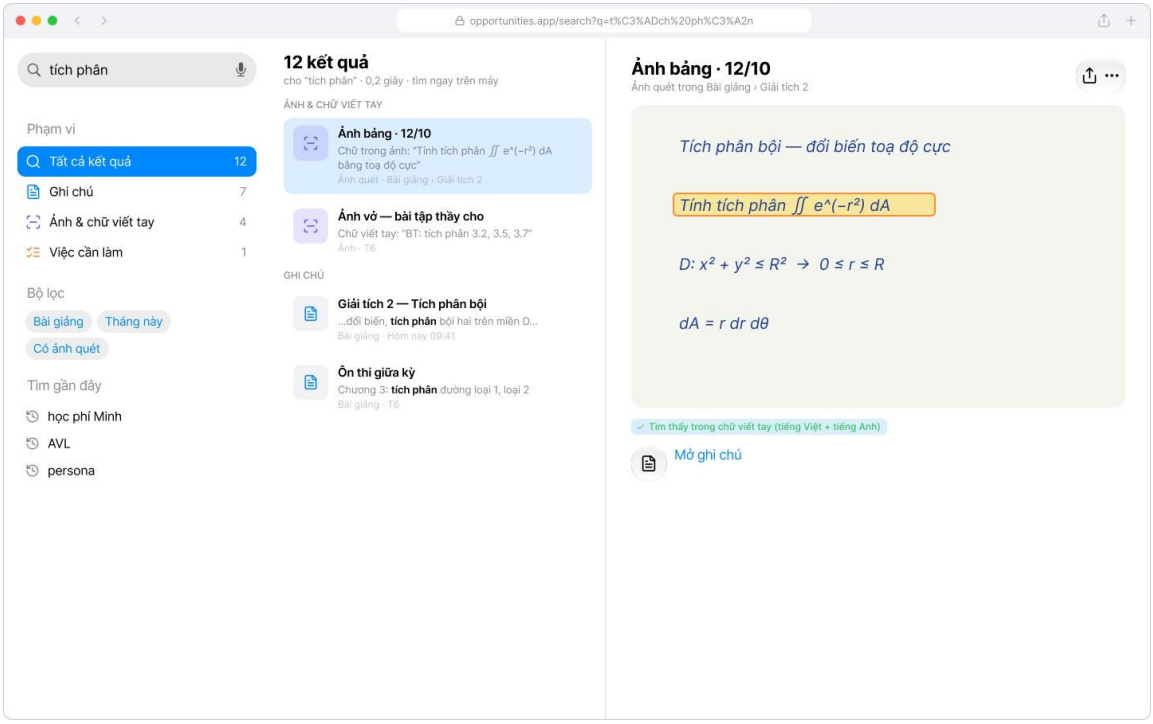


Figure 51. W04 — Search — scope filters, results, and the OCR match boxed on the photo

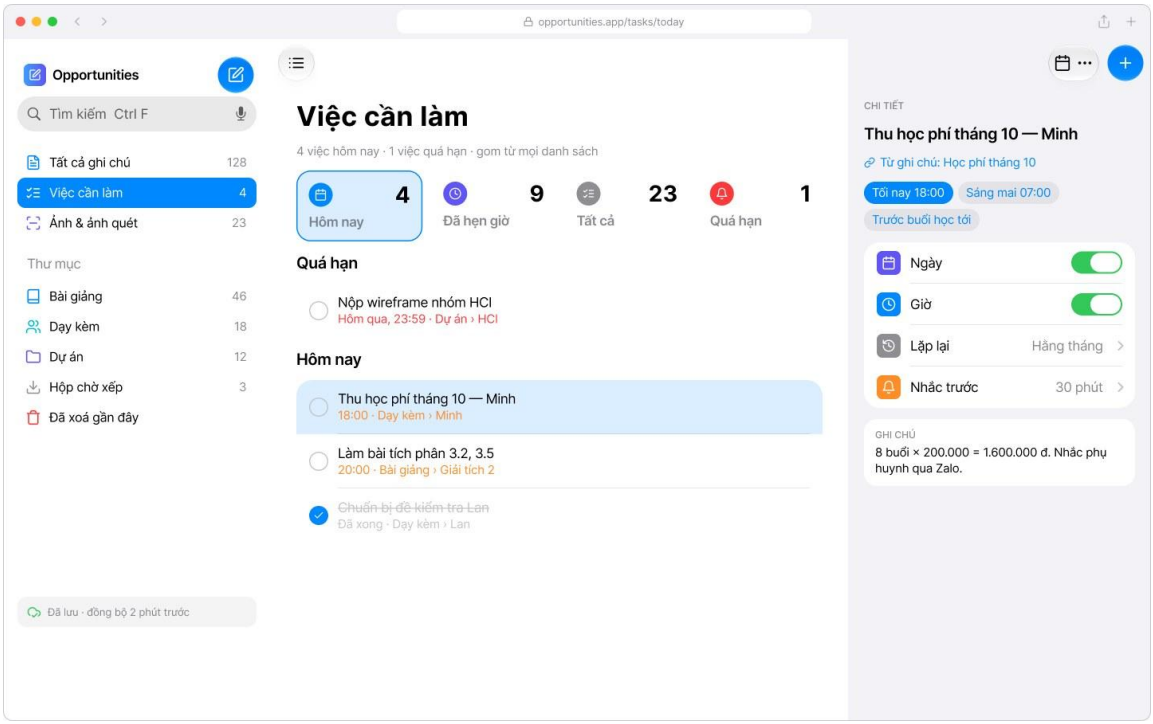


Figure 52. W05 — Tasks — smart lists and the reminder inspector

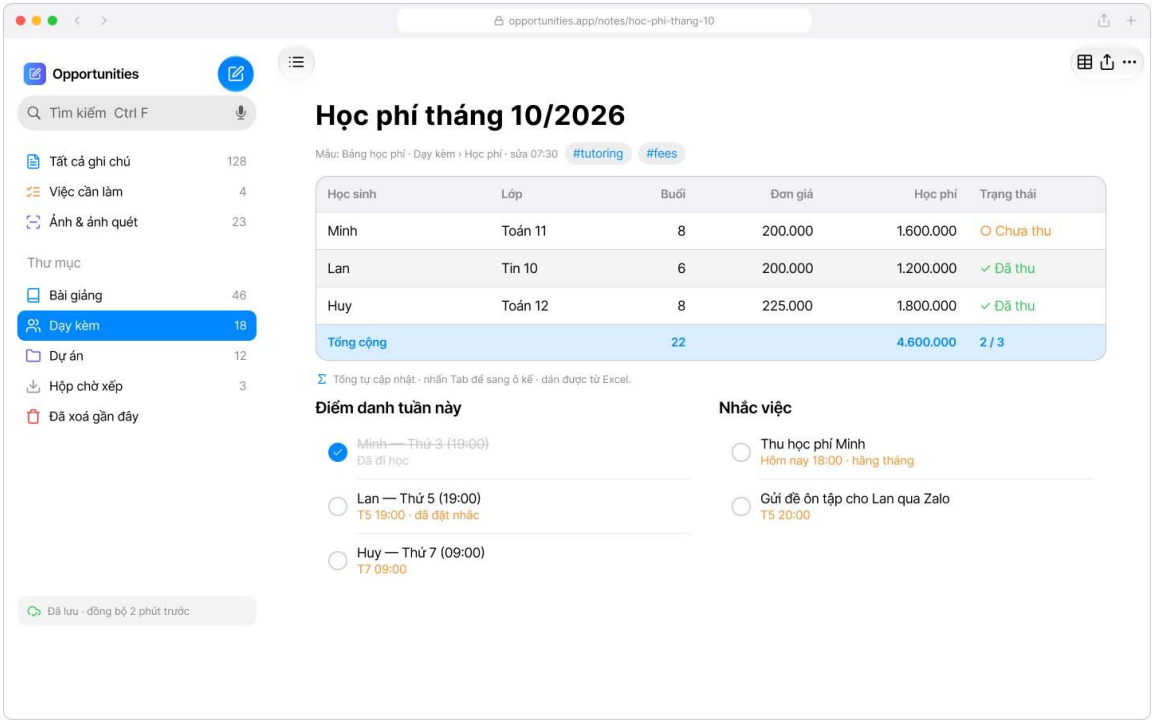


Figure 53. W06 — Tutoring — tuition table with zebra rows and automatic total, attendance

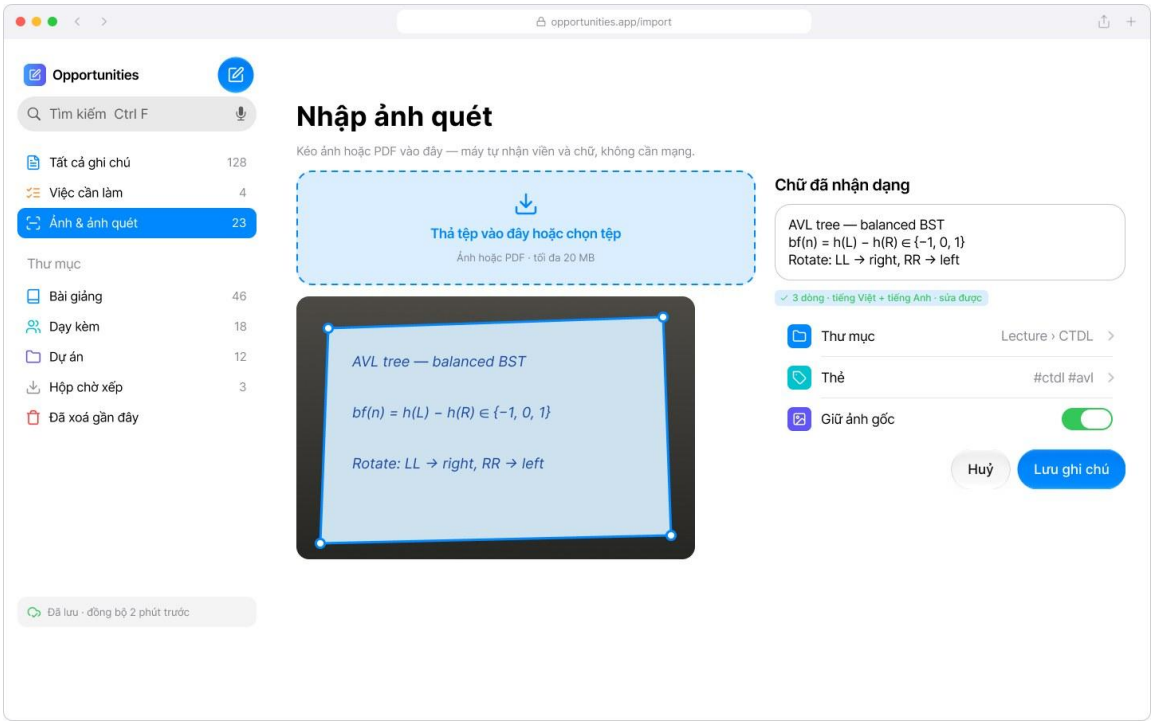


Figure 54. W07 — Import scan — drag & drop, edge correction and recognized text review

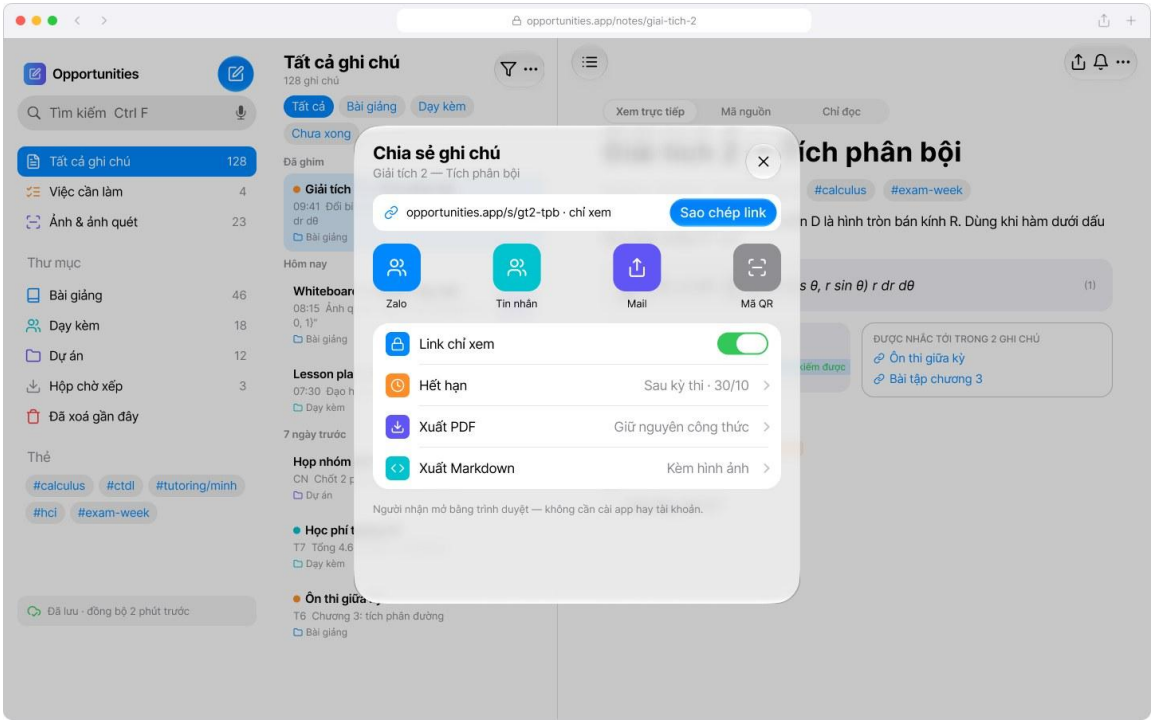


Figure 55. W08 — Share & export dialog — read-only link, expiry, PDF / Markdown

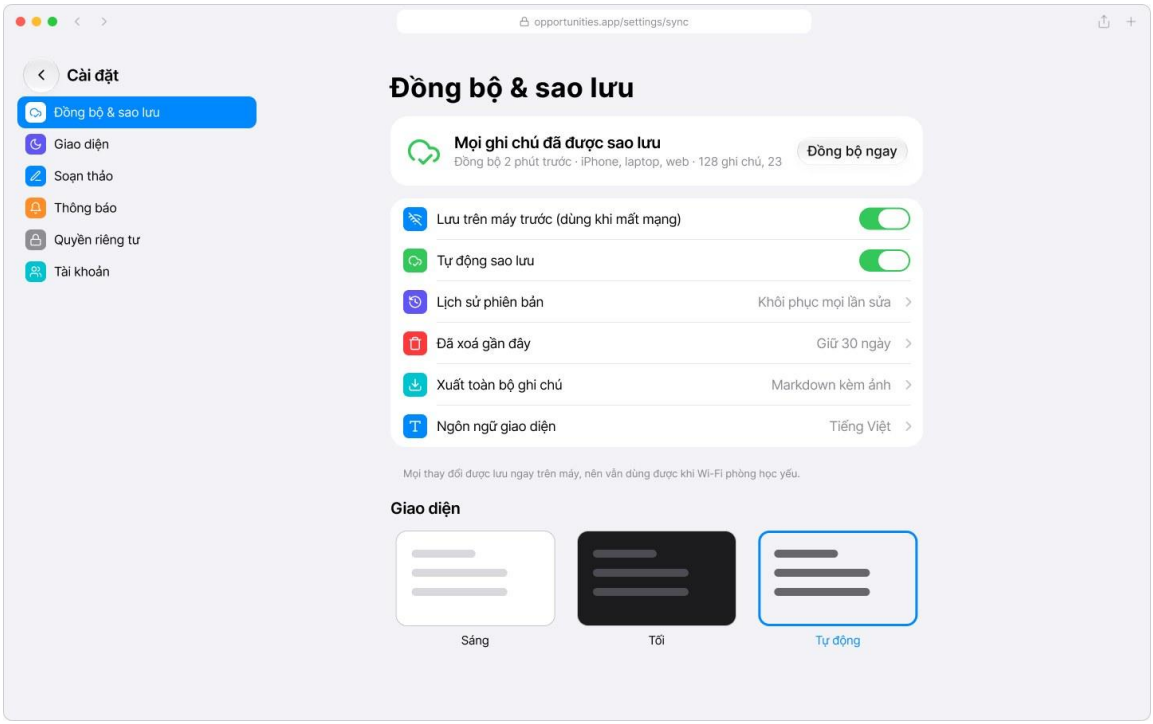


Figure 56. W09 — Settings — sync, backup, language and appearance

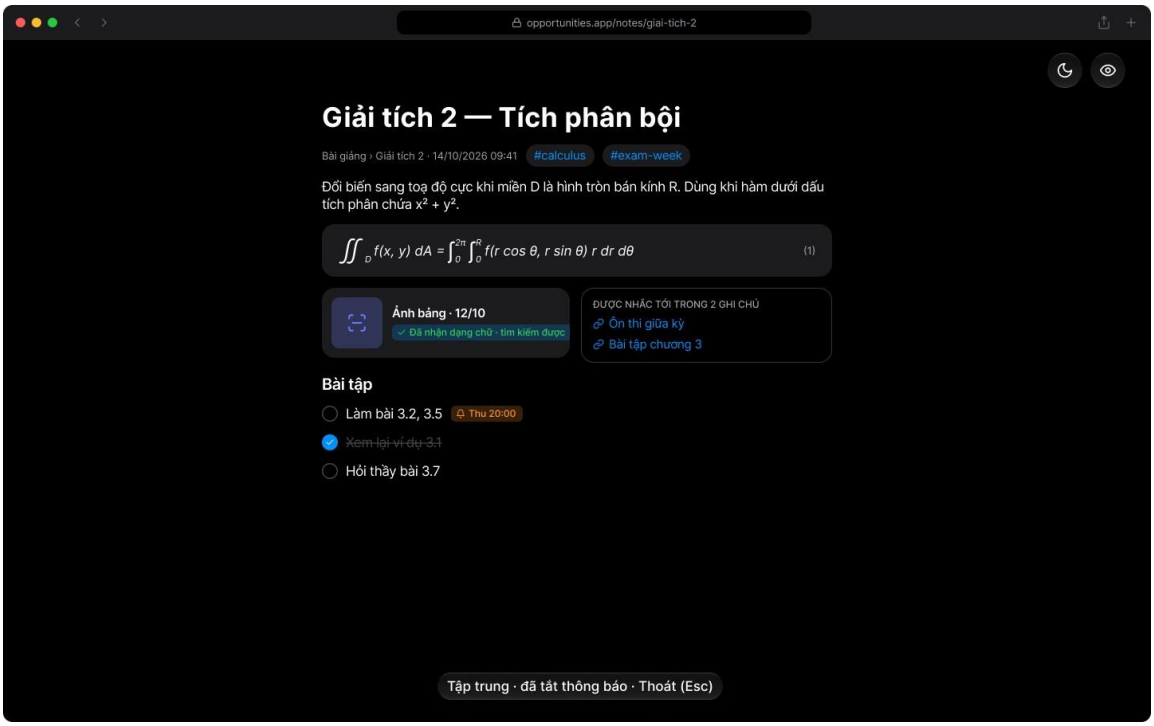


Figure 57. W10 — Dark + Focus mode for late-night revision

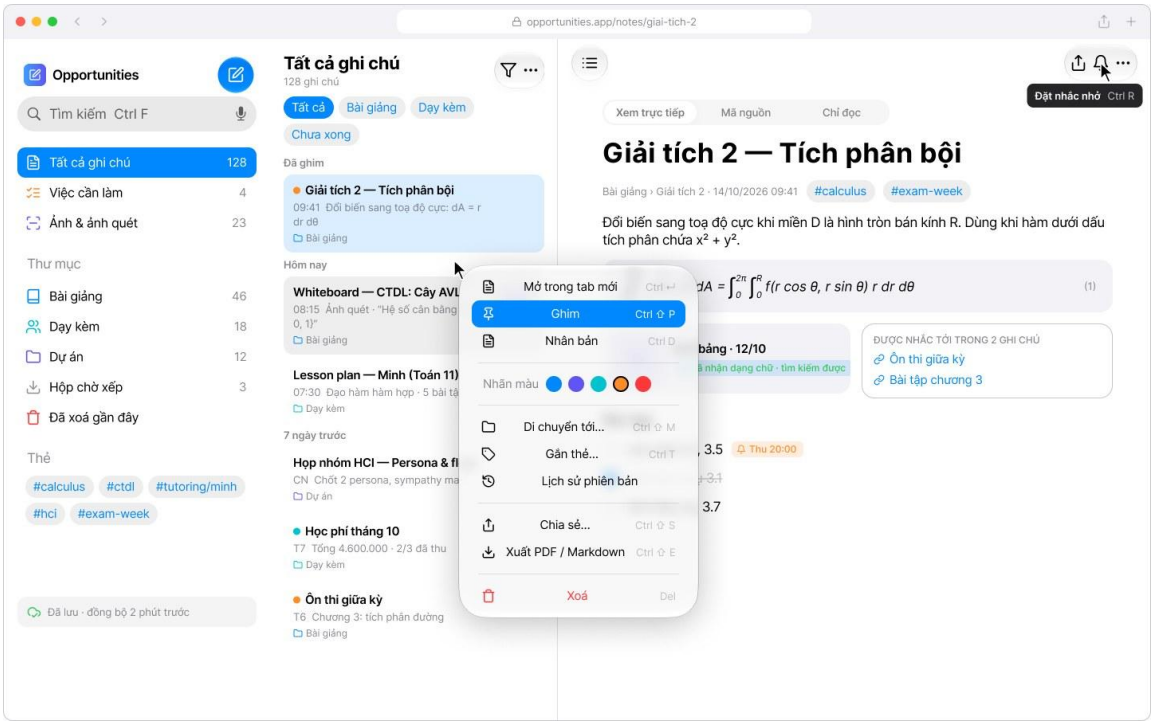


Figure 58. W11 — Right-click menu and tooltips with Ctrl shortcuts

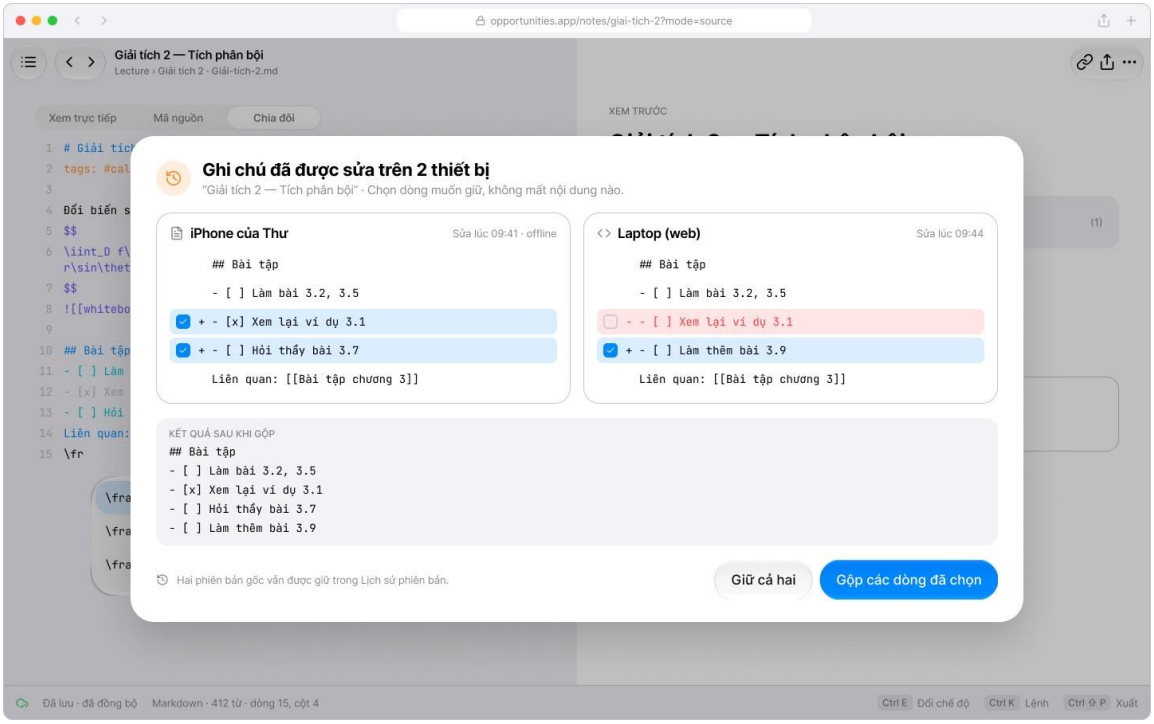


Figure 59. W12 — Sync conflict — compare and merge line by line

— THE END —